



Enhancing exchange rate forecasting with contextual sentiment indices: A fine-tuned FinBERT approach

Guoqin Gu^a , Yuping Song^{b,*}

^a School of Economics, Xiamen University, Xiamen 361005, China

^b School of Finance and Business, Shanghai Normal University, Shanghai 200234, China

HIGHLIGHTS

- A fine-tuned FinBERT outperforms with 84.33 % accuracy in forex news classification.
- Sentiment indices from FinBERT enhance exchange rate forecasting.
- Sentiment persistence and decay metrics provide deeper market insights in modeling.
- In-sample and out-of-sample analyses demonstrate the method's stability.

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ABSTRACT

The incorporation of sentiment as a predictor of exchange rate movements has long been of interest to the academic community. This paper argues that, due to the relative and interdependent nature of currency fluctuations, the sentiment contained in foreign exchange news cannot be effectively captured through current sentiment analysis tools. Instead, it requires models specifically adapted to the characteristics of each currency pair. To address this gap, we take EUR/USD pair as an example, construct a manually annotated news dataset and fine-tune the FinBERT model on an annual rolling basis to produce a series of specialized sentiment classifiers. Based on these models, we develop contextual sentiment indices and incorporate them as predictors of exchange rate changes. Our empirical findings offer several key insights. (1) The fine-tuned FinBERT models exhibit strong capability in identifying forex-specific sentiment, achieving an out-of-sample accuracy of 84.33 %, and outperforming traditional sentiment tools as well as four state-of-the-art large language models. (2) Using a doubly debiased machine learning framework, we show that the developed sentiment indices exert a statistically significant influence on exchange rate movements. (3) Compared with benchmark predictors—including technical indicators and macroeconomic variables—the inclusion of sentiment indices consistently enhances the forecasting performance of LSTM models, their variants, and a range of machine learning methods. Overall, this study contributes to both sentiment analysis and exchange rate forecasting by providing a currency-pair-specific modeling framework and demonstrating that fine-tuned language models can materially improve predictive accuracy in financial applications.

1. Introduction

As globalization progresses and international trade intensifies, accurately forecasting exchange rate fluctuations has become increasingly significant [1]. Both corporations and investors depend on reliable exchange rate predictions to guide international trade planning, risk management, and investment strategies. Similarly, government institutions utilize these forecasts to design monetary policies and address

economic challenges. However, the foreign exchange (forex or FX) market is highly intricate, influenced by shifts in national policies and economic conditions, with various factors driving exchange rate changes [2,3]. Consequently, multivariate models capable of capturing information from multiple sources are frequently employed for exchange rate forecasting.

Current multivariate forecasting models of exchange rates can be broadly categorized into three groups: structural econometric models,

* Corresponding author.

E-mail address: songyuping@shnu.edu.cn (Y. Song).

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1568-4946/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Sentiment analysis for financial market.

References	Financial market	Financial Instrument	Sentiment Analysis Methods	Text Data	Method Category	Forecasting Methods
Khadjeh Nassirtoussi et al. [22]	Foreign Exchange Market	USD/EUR	Tokenization based on Open WordNet Dictionary & Feature Selection based on TF-IDF	News Headline	Statistic-based Methods	-
Naderi Semirami et al. [28]	Foreign Exchange Market	USD/EUR, USD/JPY, USD/CHF, USD/GBP	Text Mining based on TF, TF-IDF & Sentiment Polarity Analysis based on FX-Dictionary	Economy Calendar News	Statistic-based Methods	Further Exchange Rate Forecasting based on RF, SVM, and XGB
Y. Li et al. [29]	Stock Market	Stock Price	Sentiment Classification using Naïve Bayes on Expert-Labeled Stock Message Corpus	Online Posts	Statistic-based Methods	Further Stock Price Forecasting based on LSTM
Li et al. [30]	Crude Oil Market	Crude Oil Price	Text Mining based on CNN & Feature Selection based on LDA Topic Modeling	News Headline	DL-based Methods	Further Oil Price Forecasting based on RF and SVM
Sohangir et al. [31]	Stock Market	Stock Price	Sentiment Analysis based on LSTM, Doc2vec and CNN	Opinions on StockTwits	DL-based Methods	-
Sousa et al. [32]	Stock Market	Dow Jones Industrial Index	Sentiment Analysis based on fine-tuning Bert	News Article	PM-based Methods	-

Note: If the article employs different models to further forecast financial instruments and discusses the effectiveness of sentiment based on this, the models used are listed in the last columns of the table.

time-series models, and machine learning models. Structural econometric models typically rely on macro- and micro-level predictive factors and aim to improve linear models' performance through careful selection of predictors [4]. Time-series models, such as the ARIMA and GARCH families, focus on capturing temporal dependencies and volatility patterns in historical data without necessarily incorporating economic theory explicitly [5]. Machine learning models, on the other hand, encompass a wide range of approaches. Among these, recurrent neural networks (RNNs), particularly long short-term memory (LSTM) networks, have demonstrated superior forecasting performance. Advances in technology have also led to numerous enhancements to LSTM-based forecasting models [6].

In addition to refining model architectures, extensive research has focused on improving predictors, which can be categorized into two types: intrinsic time-series predictors and external information-based predictors.

- 1. Intrinsic time-series predictors:** These rely solely on the information contained within the time series itself. The two main approaches are time-series decomposition and feature engineering. Decomposition techniques, such as ICA-LSTM [7] and EEMD-LSTM [8], derive additional modes from the original series to serve as predictors. Feature extraction methods are more diverse; for instance, Ray et al. [9] use ARIMA's forecasting residuals as LSTM predictors, while Wang et al. [10] and Liu et al. [11] apply convolutional neural networks (CNNs) to extract additional features from raw time series data for use in LSTM models. Constructing technical indicators is another intuitive feature engineering approach, with numerous studies (e.g., Hsu et al., Zarrabi et al., Panopoulou et al. [12–14]) validating its effectiveness in forex market prediction. Alonso-Monsalve et al. [15], for example, integrate technical indicators as inputs for LSTM models.
- 2. External information-based predictors:** These predictors incorporate crucial external context essential for understanding exchange rate fluctuations. Macroeconomic variables, such as interest rate differentials, inflation rates, GDP growth, and trade balances, are frequently employed, as they reflect the underlying economic conditions influencing currency values (e.g., Almeida et al., Rime et al., Alexandridis et al. [16–18]). Beyond traditional indicators, global factors like commodity prices, geopolitical events, and monetary policy shifts by major central banks have also proven valuable [19–21]. Additionally, sentiment indices derived from news, financial reports, or social media offer insights into market expectations and investor behaviour [22]. These external predictors, when

integrated into advanced models, enhance the ability to capture the complex interactions driving exchange rate movements.

Sentiment indices of news can provide valuable insights for exchange rate prediction, as news often reflects the viewpoints and emotions of opinion leaders, thereby shaping market sentiment and triggering investor reactions. As early as the early 1980s, scholars proposed news-based models for exchange rate determination [23,24]. Since then, additional econometrics empirical evidence has been presented regarding the impact mechanism of market sentiment indices on exchange rates [25–27]. Recently, several studies have attempted to predict exchange rates based on news texts (Nassirtoussi et al., Lee et al., Lv et al., Hajek et al. [20–23]). Nassirtoussi et al. developed a three-layer algorithm to convert news text into vectors. These layers include: semantic abstraction using hypernyms, sentiment-weighted calculation based on the SentiWordNet dictionary, and dimensionality reduction of features. They further used the direction of exchange rate movement over the next hour as the target, training an SVM model to predict exchange rate trends. Lee et al. employed the SnowNLP sentiment analysis tool to compute the sentiment scores of articles, which were then used to construct a sentiment index as a factor for LSTM-based prediction. Lv et al. trained a Word2Vec model to assess the sentiment orientation of vocabulary, further training an LSTM to integrate the sentiment of comment sentences. Hajek et al. fine-tuned FinBERT using over 4000 sentences from the Financial PhraseBank, labeled by financial experts to indicate the potential impact of information on stock prices, and subsequently used the fine-tuned model to determine the sentiment orientation of exchange rate-related texts. Table 1 presents sentiment analysis methods proposed in past research for different financial market analysis targets. If sentiment was used as a predictor, the table also shows the prediction models employed in these studies.

Existing research on sentiment-based exchange rate forecasting exhibits two major limitations.

First, while many studies focus on developing sophisticated models to extract sentiment and apply it to downstream tasks, they often overlook a more fundamental question: What constitutes investor sentiment, and how does it originate? Volatility is essentially a macro-level manifestation of investors' trading behaviour, which is in turn heavily driven by sentiment. The widely adopted GARCH model shows that volatility evolves along three key dimensions: magnitude, rate of change, and persistence. Given that investor sentiment underlies trading dynamics, and trading behaviour directly shapes price volatility, it follows that these three volatility dimensions reflect the underlying dynamics of collective sentiment. Applying the three dimensions of sentiment to concrete forecasting tasks is a promising approach.

Second, a simple yet often overlooked fact is that sentiment classification for foreign exchange news differs fundamentally from that in other domains, an issue rarely addressed in existing literature. For instance, consider the EUR/USD currency pair and two pieces of news:

1. *The US dollar is weakening significantly!*
2. *There is strong confidence in the strength of the euro!*

Conventional sentiment analysis tools would classify the first as “Negative” and the second as “Positive.” However, in the context of EUR/USD, both indicate a strengthening of the euro and thus should be interpreted as “Positive”. The bidirectional nature of exchange rate movements means sentiment must be classified in a currency-pair-specific context. This can only be achieved through models fine-tuned on foreign exchange news tailored to specific currency pairs. As noted, Lee and Hajek applied general-purpose sentiment libraries and stock-market-tuned models to forex news, resulting in misaligned sentiment interpretations. While Nassirtoussi and Lv trained models on forex-specific texts, their objective was to predict exchange rate trends, not to preserve the original semantic meaning of the text. Thus, there remains a significant research gap in developing sentiment classification models that accurately reflect the true semantic orientation of forex news.

Building on these two limitations, this paper argues that deconstructing and verifying sentiment components not only improves downstream forecasting accuracy but also provides novel insights for exchange rate theory. We emphasize the importance of comparing sentiment with other predictors, compare sentiment across its three dimensions, and conduct both in-sample analysis and out-of-sample forecasting to evaluate these relationships.

Accordingly, we propose a finetuned FinBERT called FXFinBERT, a sentiment classification model tailored to specific currency pairs. Using this model, we construct a comprehensive pool of sentiment factors aligned with the three dimensions of volatility implied by the GARCH framework. Specifically, we collected 15 years of forex news data and manually annotated approximately 10 % of it to train our model. FXFinBERT outperforms TF-IDF and four advanced large language models in classification accuracy. Finally, we demonstrate—via in-sample doubly debiased machine learning (DDML) regression and out-of-sample multi-models forecasting—that our sentiment dimensions significantly enhance exchange rate predictability. Our framework can also be extended to interval forecasting and other predictive modeling tasks.

The main contributions of this paper are as follows:

1. Dynamic features framework of sentiment: We move from GARCH-driven insights to sentiment design, by measuring investor sentiment along the same three axes—intensity, change, and persistence—we can more accurately capture the evolving nature of market sentiment. This motivates the construction of a three-dimensional sentiment index pool to better characterize sentiment dynamics and improve volatility forecasting performance.

2. A currency-pair-specific sentiment classification model: We develop FXFinBERT, a sentiment classification model specifically fine-tuned on forex-related news, to address the limitations of general-purpose sentiment tools in the foreign exchange domain. Through a rolling fine-tuning scheme over 15 years of historical data, FXFinBERT achieves robust out-of-sample performance, with consistent improvements in F1-score and accuracy across multiple years. Compared to existing approaches—including zero-shot FinBERT, TF-IDF, and proprietary LLMs such as GPT-4, Claude-3.5-sonnet, Qwen-plus, and GLM-4—FXFinBERT achieves SOTA accuracy (84.33 %) and macro F1-score (84.60 %).

3. New empirical evidence on sentiment impact: Our approach reveals key differences between sentiment dimensions and sentiment measures constructed by various models through in-sample doubly debiased machine learning (DDML) analysis. Furthermore, out-of-

sample forecasting results demonstrate the effectiveness of sentiment features in enhancing the predictive performance of a wide range of models, providing novel empirical support for sentiment-driven theories in the foreign exchange market.

The remainder of this paper is structured as follows: [Section 2](#) outlines the research methodologies utilized in this study. [Section 3](#) details the data sources and predictors. [Section 4](#) begins with the research design, providing an in-depth discussion of the need for fine-tuning and the process and results of fine-tuning FXFinBERT. It then presents in-sample analysis based on double machine learning and a series of out-of-sample predictions and comparisons, primarily focusing on LSTM. [Section 5](#) summarizes the study and provides future outlook.

2. Methodology

2.1. Research methods overview

This section provides an overview of the research methods to facilitate a comprehensive understanding of the methodology. Our study comprises three core modules: (1) sentiment construction based on forex news, (2) exchange rate forecasting using various machine learning methods, and (3) performance evaluation and interpretation.

2.2. Sentiment construction

2.2.1. TF-IDF

The Term Frequency-Inverse Document Frequency (TF-IDF) model was used as a baseline for sentiment analysis in this study. After pre-processing, we transformed the news into numerical vectors using a TF-IDF vectorizer. These features were then used to train a SVM classifier, which categorized sentiments into Positive, Neutral, and Negative. The model was trained using the same rolling window approach from 2016 to 2023.

2.2.2. FinBERT and FXFinBERT

FinBERT is a specialized version of BERT, designed for applications within the financial domain by Araci [33]. It adapts BERT's architecture for financial texts using Transformer encoders. In this study, we leveraged the FinBERT model developed by Huang et al. [34]. According to Huang, FinBERT was pre-trained on a vast corpus of financial documents, including corporate reports, financial analyst reports, and earnings call transcripts, totaling 4.9 billion tokens. This corpus is 50 % larger than the 3.3 billion tokens used by Google for pretraining BERT, equipping FinBERT with a deep understanding of financial terminology not commonly found in general texts.

We fine-tuned FinBERT on an NVIDIA 3090 GPU to develop a sentiment analysis model specifically tailored for the foreign exchange market, with a particular focus on EUR/USD sentiment, named FXFinBERT. A custom dataset designed for this domain was employed during fine-tuning to improve predictive accuracy. This approach capitalizes on FinBERT's advanced capabilities in processing financial texts, enabling more precise sentiment identification within the context of the foreign exchange market. In the following sections, we illustrate the necessity and advantages of fine-tuning FinBERT, as opposed to alternative methods, for conducting sentiment analysis of foreign exchange sector news.

2.2.3. LLMs

Large Language Models (LLMs), such as GPT-4, have demonstrated significant advancements in natural language understanding and generation, including applications in sentiment analysis. These models are pre-trained on extensive datasets comprising diverse textual sources, enabling them to capture intricate language patterns and contextual nuances. In this study, LLMs serve as a comparison against our fine-tuned FXFinBERT. While LLMs offer impressive out-of-the-box performance, their lack of specialized training in the foreign exchange domain

may limit their accuracy in handling nuanced forex-related sentiment classifications.

2.2.4. Indicator selection methods

To identify the most informative sentiment indices, we implemented a multi-stage feature selection framework combining correlation filtering and ensemble-based importance scoring. Initially, we applied Pearson correlation and Granger causality tests [35] to eliminate indices with high redundancy, removing those with pairwise correlations exceeding 0.95. Subsequently, we employed five complementary feature selection techniques to assess the predictive value of each remaining index: LightGBM importance, XGBoost importance, Random Forest feature importance, Recursive Feature Elimination with Cross-Validation (RFECV) [36,37], and the Boruta algorithm [38]. These methods jointly evaluate both model-based importance scores and ranking-based relevance. To synthesize these results, we normalized the outputs of each method and computed a weighted average to derive a composite Sentiment Value Indicator (SVI) for each index. Indices ranking in the bottom 25 % of SVI scores were excluded from further analysis.

For the normalization of indicators, the formula is:

$$\text{Norm - Indicator} = \frac{\text{Indicator Value} - \min(\text{Indicator})}{\max(\text{Indicator}) - \min(\text{Indicator})} \quad (1)$$

For the normalization of rankings, the formula is:

$$\text{Norm - Ranking} = 1 - \frac{\text{Ranking Value} - 1}{\max(\text{Ranking}) - 1} \quad (2)$$

$$\text{SVI} = \sum_{i=1}^3 w_i \cdot \text{Norm - Indicator} + \sum_{i=4}^5 w_i \cdot \text{Norm - Ranking\#} \quad (3)$$

2.3. Exchange rate forecasting

2.3.1. Doubly debiased machine learning

Classical regression methods are often reliant on model assumptions, which may not align well with the complexities of the real world. Additionally, they lack a systematic mechanism for handling high-dimensional data. The Doubly Debiased Machine Learning (DDML) method addresses these issues by using machine learning models to estimate covariate adjustment functions, making it robust to misspecification of functional forms and overfitting. Moreover, it allows for the estimation of average treatment effects through orthogonal regression.

In this study, we aim to evaluate the impact of sentiment indices (treatment variables denoted by s) on future exchange rate changes (outcome variables denoted by Δ). Given the complex influence mechanisms of multiple macroeconomic and microeconomic variables in the exchange rate market, DDML is better suited for our analysis than classical regression methods. We consider the following model:

$$\Delta_{t,k} = \theta_0 S_t + g_0(X_t) + \epsilon_t \# \quad (1)$$

$$S_t = m_0(X_t) + \zeta_t \# \quad (2)$$

$$\Delta_{t,k} = \text{Close}_{t+k} - \text{Close}_t \# \quad (3)$$

In the above model, time is measured at a daily frequency, denoted by t , and $\Delta_{t,k}$ represents the exchange rate change at time t over the next k days. The vector θ_0 is the set of regression coefficients we aim to estimate. The unknown function $g_0(X_t)$ captures the effect of the covariates on the outcome variable, while $m_0(X_t)$ reflects the potential influence of covariates on the treatment variable. The error terms in the model are denoted by ϵ_t and ζ_t .

We analyze the impact of several key sentiment indices, as described in the subsequent section, and examine their effects on exchange rate movements over 1, 3, 7, 30, and 90 days. The covariates used in the functions $g_0(X_t)$ and $m_0(X_t)$ are the technical and macroeconomic in-

dicators discussed in the next section.

2.3.2. LSTM

Long Short-Term Memory (LSTM) networks are a specialized kind of Recurrent Neural Network (RNN) designed to address the problem of learning long-term dependencies. Traditional RNNs struggle to capture long-distance relationships within data due to issues like vanishing or exploding gradients [39]. LSTMs overcome these challenges with a unique architecture that allows them to retain information over longer periods effectively due to Hochreiter and Schmidhuber [40].

The key innovation in LSTMs is the introduction of a memory cell and three gate mechanisms: the input gate, the forget gate, and the output gate. These components work together to regulate the flow of information into and out of the cell, allowing the network to make informed decisions about which data to store, discard, or pass through at each step in a sequence.

The LSTM updates its internal states through a series of operations involving these gates and the memory cell, as outlined in the following equations:

1. Forget gate activation:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

This step involves the sigmoid function σ , which outputs values between 0 and 1, indicating how much of the past information to keep from the memory cell.

2. Input gate activation:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

Simultaneously, another sigmoid function decides which values will be updated.

3. Candidate memory cell:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (6)$$

A tanh layer creates a vector of new candidate values that could be added to the state.

4. Memory cell update:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (7)$$

The old memory cell is updated to the new cell state, combining past information (gated by the forget gate) and new candidate values (gated by the input gate).

5. Output gate activation:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8)$$

This gate controls the extent to which the memory cell's information is used to compute the output.

6. Final output:

$$h_t = o_t * \tanh(C_t) \quad (9)$$

The output of the LSTM unit is filtered by the output gate, ensuring that only the relevant information is passed through.

By selectively updating and discarding information, LSTMs can model complex sequences with long-range temporal dependencies, making them highly effective for foreign exchange rate time-series forecasting. In this paper, we use the PyTorch framework to build our neural network. The three-dimensional tensor input for LSTM and the neural network structure defined in this paper are shown in Fig. 22.

2.3.3. Other baseline models

To provide a comprehensive performance benchmark, we include a range of baseline models. For traditional machine learning, we employ Support Vector (SVR), Random Forest (RFR), Gradient Boosting (GBR), LightGBM (LGBM), Elastic Net (ENR), and K-Nearest Neighbors (KNN). For LSTM variants, we consider CNN-LSTM [11], Bidirectional LSTM (BiLSTM) [41], and Gated Recurrent Units (GRU) [42]. We also include time series models ARIMA and GARCH.

2.4. Model evaluation & interpretation

2.4.1. Evaluation metric

In this study, we employed five loss functions to evaluate the performance of each model. These include Mean Absolute Error (MAE), Mean Squared Error (MSE), Mean Squared Logarithmic Error (MSLE), Mean Absolute Percentage Error (MAPE), and Logarithm of the Hyperbolic Cosine Loss (LogCosh), each defined by the following formulas:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (11)$$

$$\text{MSLE} = \frac{1}{n} \sum_{i=1}^n (\log(1 + y_i) - \log(1 + \hat{y}_i))^2 \quad (12)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (13)$$

$$\text{LogCosh} = \frac{1}{n} \sum_{i=1}^n \log(\cosh(\hat{y}_i - y_i)) \quad (14)$$

Each function offers a different perspective on error measurement, ranging from absolute errors (MAE), squared errors sensitive to large deviations (MSE), logarithmic transformations suitable for exponential data (MSLE), relative percentage errors (MAPE), to smooth and stable error gradients (LogCosh). Collectively, these metrics allow us to assess and compare the predictive capabilities and robustness of different models under varied conditions and scales, ensuring a thorough validation of their effectiveness in forecasting exchange rates.

2.4.2. Diebold-Mariano test

The Diebold-Mariano (DM) test evaluates the statistical significance of the difference in predictive accuracy between two forecasting models [43]. Given two sets of prediction errors, E_a and E_b , for models A and B over T periods, the differences in errors are calculated as $d_i = a_i - b_i$. The test then computes the mean and standard deviation of these differences:

$$d_{\text{mean}} = \frac{\sum_{i=1}^T d_i}{T} \quad (15)$$

$$d_{\text{std}} = \sqrt{\frac{\sum_{i=1}^T (d_i - d_{\text{mean}})^2}{T - 1}} \quad (16)$$

The DM statistic is:

$$\text{DM} = \frac{d_{\text{mean}}}{d_{\text{std}}} \quad (17)$$

This statistic follows a standard normal distribution under the null hypothesis that both models have equivalent predictive accuracy. The significance of the DM statistic can be assessed using standard normal distribution critical values to determine if there is a significant difference in performance between the two models. Therefore, the DM test will be used to assess whether there is a statistical difference in the

predictive effect of the model before and after the inclusion of the news indices.

2.4.3. Permutation importance

To evaluate the importance of the sentiment index in our model, we employed the permutation feature importance method. This approach quantifies a feature's significance by observing changes in model performance when the feature's values are randomly shuffled, thereby disrupting its relationship with the target variable [44]. Our procedure follows the algorithm proposed by Fisher et al. [45] and Qian et al. [46] which is a model-agnostic technique:

1. Given a trained model $\hat{f}(X)$ and loss function $L(y, \hat{f}(X))$, compute the original test loss value $e_{\text{orig}} = L(y, \hat{f}(X))$.

2. Randomly permute the selected feature x 100 times within the test dataset, calculate the test loss each time as e_{perm} , and compute the ratio $\frac{e_{\text{perm}}}{e_{\text{orig}}}$ to measure the feature's importance.

In our analysis, the degree to which the performance declined after permuting the sentiment index indicated its importance—greater performance drops corresponded to higher importance, signifying that the model heavily relied on the sentiment index for accurate predictions.

3. Data

Our sample spans the period from January 2009 to December 2023 on a daily frequency (5476 observations). The primary focus is on the exchange rate of the EUR/USD currency pair, with data sourced from *finance.yahoo.com*. Missing values in the dataset were addressed using linear interpolation.

Following the conceptual framework discussed in the Introduction, our analysis is grounded in two complementary perspectives: Intrinsic Time-Series Predictors and External Information-Based Predictors. Accordingly, we aim to examine the extent to which sentiment indicators can improve existing forecasting methods on a marginal basis. Previous studies, such as those by Nabi et al. [47] and Ntakaris et al. [48], have demonstrated the effectiveness of certain technical indicators in time-series forecasting. These indicators can be viewed as a complement to the intrinsic information contained within the time series itself. Therefore, predictions incorporating these technical indicators will serve as a baseline for comparison. Another major factor influencing exchange rates is the macroeconomic environment. For our primary research subject, the EUR/USD currency pair, it is meaningful to incorporate macroeconomic variables from both the Eurozone and the United States. Previous studies, such as those by Faust et al. [49], Wang et al. [50], and Alexandridis et al. [18], have demonstrated the impact of macroeconomic variables on exchange rates. Therefore, macroeconomic indicators will also serve as a benchmark for comparison. In the following subsections, we introduce the specific variables used under each predictor group. The third subsection details our approach to extracting sentiment-based predictors from news text, which serve as the key innovation of this study.

3.1. Technical indicators as predictors

Specifically, we selected 20 commonly used technical indicators for exchange rate forecasting, categorizing them into three groups: (1) intra-day indicators (measuring the magnitude of exchange rate fluctuations from the opening to the closing of a single trading day), (2) inter-day indicators (capturing exchange rate movements over 1–10 days), and (3) amplitude indicators (quantifying the volatility and rate of change in exchange rates). A detailed list of these technical indicators is presented in Table 10, with their mathematical formulations available in the studies by Agrawal et al. [51] and Gu et al. [52].

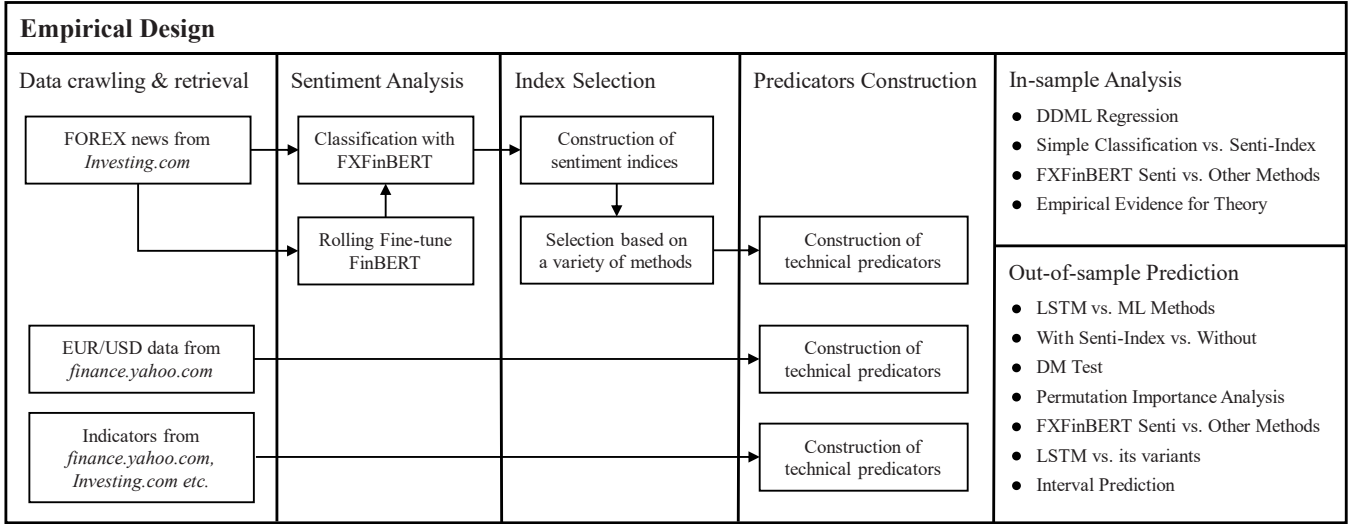


Fig. 1. Empirical Design.

3.2. Macroeconomic indicators as predictors

We selected 20 commonly used macroeconomic indicators for exchange rate forecasting, taking into account the availability of daily data. A detailed list of these indicators is presented in Table 11, and is briefly described below.

The first set of predictors relates to the equity markets in the Eurozone and the United States. Studies by Andreou et al. [53] and Wang et al. [54] highlight the close relationship between equity and foreign exchange markets. For the U.S. stock market, we use the S&P 500 Index (GSPC) and the NASDAQ Composite Index (IXIC). For the European stock market, we include the Euronext 100 Index (N100), which reflects the performance of the 100 largest companies listed on the Euronext exchange, and the EURO STOXX 50 Index (STOXX50E), a blue-chip index tracking the 50 largest companies in the Eurozone.

The second set of predictors pertains to the commodity markets. According to Liu et al. [55], risk sentiment often manifests in the price movements of safe-haven assets such as gold (GOLD) and West Texas Intermediate crude oil futures (COWTIF). Additionally, Bitcoin (BIT) is included as a measure of new risk assets. To capture market panic, we employ the CBOE Volatility Index (VIX) and the Thomson Reuters/CoreCommodity CRB Total Return Index (TRCCRBTR), which reflect global commodity market volatility and returns, respectively.

The third set of predictors is associated with bond markets. Andersen et al. [56] have shown that bond and foreign exchange markets often respond similarly to macroeconomic changes, motivating the inclusion of bond-related variables. For the U.S. bond market, we incorporate the 13-week Treasury Bill Yield (IRX), the 5-year Treasury Yield (FVX), and the 30-year Treasury Yield (TYX), representing short-term, medium-term, and long-term interest rates. For the Eurozone bond market, we include the Thomson Reuters Eurozone Government Benchmark for 2-year (TRXVEMGOV2E), 5-year (TRXVEMGOV5E), and 30-year (TRXVEMGOV30E) maturities. Following Buncic et al. [57], we also calculate the difference between the 3-Month LIBOR and the 3-Month Treasury Bill Secondary Market Discount Rate (LTS).

The fourth set of predictors captures inflation dynamics. We include the year-over-year (USCPIY2Y) and month-over-month (USCPIM2M) changes in the U.S. Consumer Price Index (CPI), as well as the corresponding year-over-year (EuroCPIY2Y) and month-over-month (EuroCPIM2M) changes in the Eurozone CPI.

3.3. Sentiment as predictors

Investing.com, established in 2007, is recognized as one of the leading

financial platforms and news websites globally. The Foreign Exchange (Forex) section of this platform provides comprehensive coverage of forex-related news, serving as a crucial resource for market participants.

In this study, we collected and processed a total of 74,537 news headlines accompanied by brief summaries from *Investing.com*, spanning the period from January 2009 to December 2023. To fine-tune the pre-trained model in subsequent steps, we randomly sampled 10 % of the data from each year, constructing a dataset containing 7455 news. These news texts were analysed to assess their impact on the EUR/USD exchange rate trends, with each piece labelled as Positive, Negative, or Neutral.

Here, "Positive" news is defined as having a driving effect on increasing the EUR/USD exchange rate, i.e., favourable for the euro or unfavourable for the dollar. Conversely, "Negative" news is defined as having a driving effect on decreasing the EUR/USD exchange rate, i.e., unfavourable for the euro or favourable for the dollar. "Neutral" news refers to news with no discernible driving impact on the EUR/USD exchange rate.

News related to exchange rates differs from general news because, in the context of exchange rate movements, both "Positive" news indicating favourable conditions for the euro and traditionally "Negative" news indicating unfavourable conditions for the dollar typically contribute to an increase in the EUR/USD exchange rate and should therefore both be classified as "Positive." This categorization approach ensures scientific rigor, enabling the sentiment indices derived from these classifications to more effectively enhance exchange rate forecasting.

To improve model robustness during fine-tuning, we anticipated that it would capture patterns relatively stable across years. For instance, phrases like "Investors took profit on the US dollar," "Risk appetite increased," and "Dragged down the dollar, bearish for the dollar" often indicate a favorable outlook for the euro, with these patterns remaining consistent over time. Additionally, the model was expected to recognize the potential effects of certain keywords. For example, the release of "economic data" often influences exchange rate trends, and signals from the Federal Reserve regarding "interest rate hikes" also play a critical role. To support this, we constructed a supplementary dataset comprising 120 well-labeled sentences, evenly distributed across Positive, Negative, and Neutral labels. These sentences were crafted based on prior knowledge of exchange rate dynamics and used to enhance the rolling fine-tuning process of the pre-trained model.

Finally, the full dataset of 74,537 news was classified using a fine-tuned version of FinBERT called FXFinBERT. The resulting sentiment indices was then incorporated as a predictor in the exchange rate forecasting model.

Table 2

Classification performance report of LLMs.

Method	Macro P. ^{a)}	Macro R. ^{b)}	Macro F. ^{c)}	Balanced Acc. ^{d)}
NoFT-FinBERT	38.44 %	38.36 %	38.20 %	38.36 %
TF-IDF	49.60 %	43.81 %	39.91 %	43.81 %
GPT-4	67.30 %	64.01 %	61.32 %	64.01 %
Claude-sonnet	67.90 %	67.44 %	66.13 %	67.44 %
Qwen-plus	77.23 %	68.81 %	71.00 %	68.81 %
GLM-4-plus	72.20 %	58.90 %	55.29 %	58.90 %

Note:

- a) The macro-averaged Precision: The average precision score calculated across all classes.
- b) The macro-averaged Recall: The average recall score calculated across all classes.
- c) The macro-averaged F-score: The harmonic mean of macro-averaged precision and macro-averaged recall.
- d) Balanced Accuracy: The average of recall for each class, providing a measure that accounts for class imbalance.

4. Empirical study

The primary aim of our empirical study is to construct sentiment indices using FXFinBERT, validate their ability to enhance exchange rate prediction accuracy, and provide new evidence for existing research on exchange rates. This section begins with an outline of the study design.

4.1. Design of empirical study

Our empirical study comprises three core components: training FXFinBERT and constructing sentiment indices, conducting in-sample analysis and interpreting the economic implications of the sentiment indices, and performing out-of-sample prediction and comparing various forecasting methods. The flow of the empirical experiment is illustrated in Fig. 1.

• FXFinBERT and sentiment index

Subsection 4.2 first elaborates on why we adopted a fine-tuning approach to develop FXFinBERT for classifying exchange rate news sentiments, rather than directly leveraging pre-trained large language models (detailed in Section 4.2.1). Subsequently, we explain how sentiment indices are constructed without information leakage within historical windows, ensuring the robustness of our subsequent evaluations (detailed in Section 4.2.2). We then describe how the classification results of exchange rate news were processed, supported by intuitive evidence of their significant impact on exchange rate prediction (detailed in Section 4.2.3). Finally, a comprehensive process for constructing and selecting sentiment indices is presented (detailed in Section 4.2.4).

• In-sample analysis

Subsection 4.3 employs Doubly Debiased Machine Learning (DDML) for in-sample analysis. We first analysis that sentiment indices have a more pronounced impact on exchange rate predictions from 1 to 90 days ahead compared to raw classification results (detailed in Section 4.3.1). Then, theoretical mechanisms proposed by prior research on the influence of exchange rates are empirically supported by our results (detailed in Section 4.3.2).

• Out-of-sample prediction

Subsection 4.4 begins with a comparison showing that incorporating our constructed sentiment indices improves the predictive performance of various models (detailed in Section 4.4.1). Robustness checks and an analysis of indicator importance follow (detailed in Section 4.4.2). Next, we compare sentiment indices derived from different sentiment analysis methods, further demonstrating the effectiveness of the FXFinBERT approach (detailed in Section 4.4.3). Finally, we propose a relatively optimal predictive model building on prior research (detailed in Section

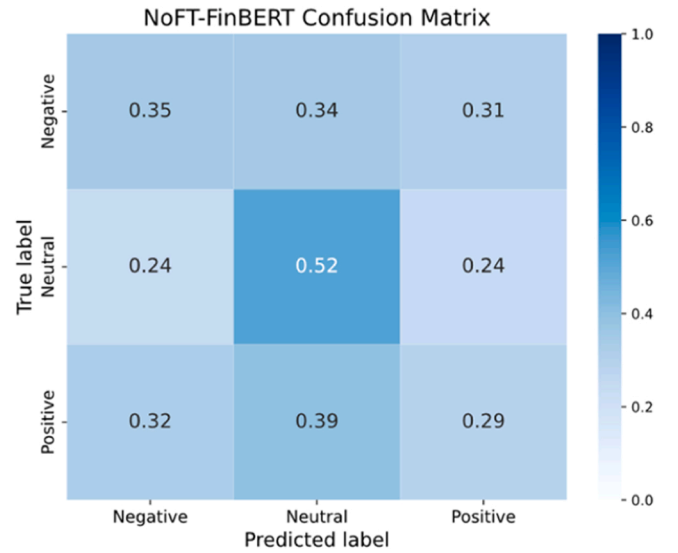


Fig. 2. Confusion Matrix of NoFT-FinBERT.

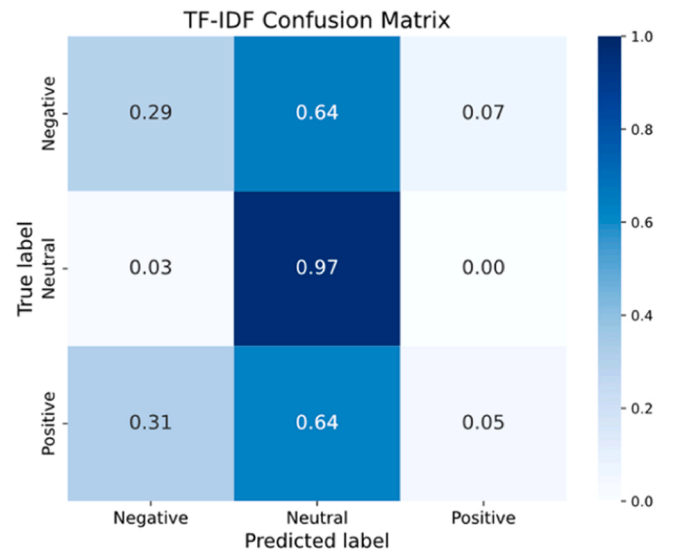


Fig. 3. Confusion Matrix of TF-IDF Method.

4.4.3).

The experiments were implemented in Python, with the computational environment comprising an Intel(R) Xeon(R) Platinum 8362 CPU and an NVIDIA RTX 3090 GPU. The software stack included Python 3.12 and PyTorch 2.3.0 for deep learning model training.

4.2. FXFinBERT and sentiment index

4.2.1. Why we need finetune?

We evaluated the classification performance of FinBERT without fine-tune (NoFT-FinBERT), the TF-IDF method, GPT-4, Claude-3.5-sonnet, Qwen-plus and Glm-4-plus on the manually labelled dataset of 7455 news. The results, presented in Table 2, raise an intriguing question: Are these results surprising? Across all models, the classification performance is suboptimal. Could this be attributed to the unique classification scheme of forex-related news? To address this, we need to conduct a more detailed examination of each model's performance.

Fig. 2 illustrates the confusion matrix for NoFT-FinBERT's classification outcomes. The results reveal a significant challenge: the model tends to misclassify news almost equally between the "Positive" and

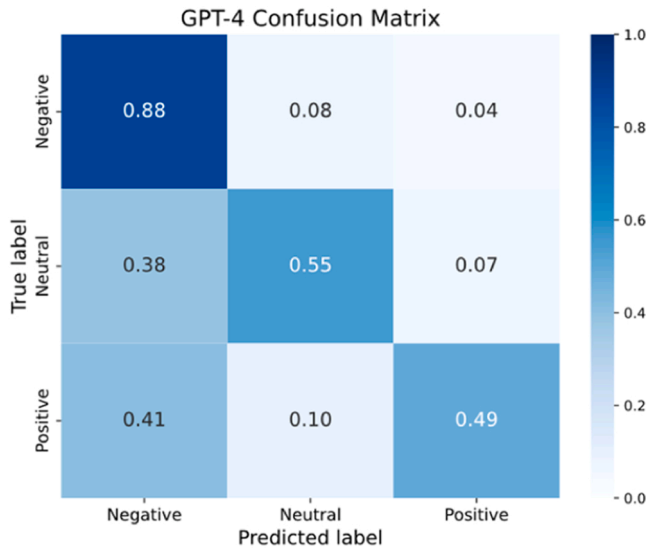


Fig. 4. Confusion Matrix of GPT-4.

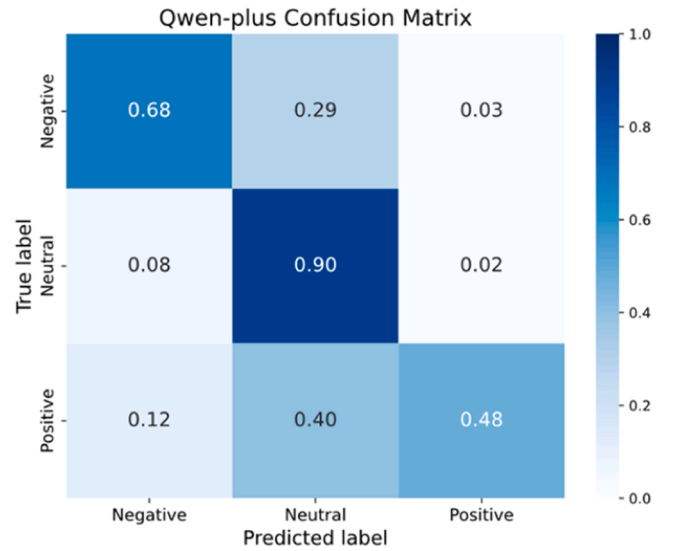


Fig. 6. Confusion Matrix of Qwen-plus.

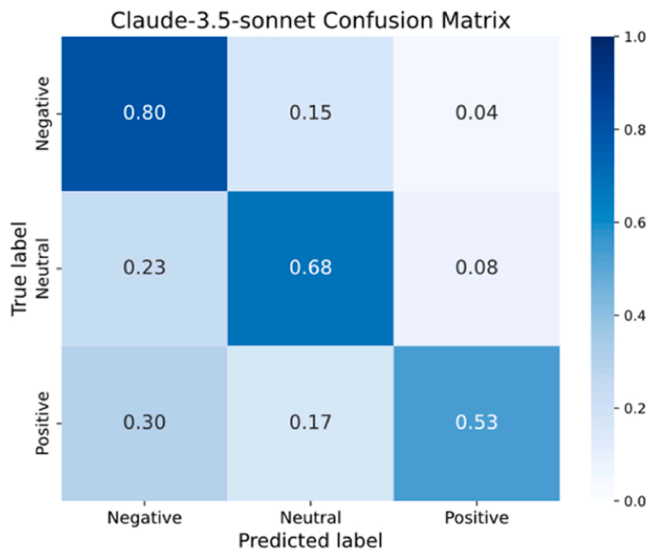


Fig. 5. Confusion Matrix of Claude-3.5-sonnet.

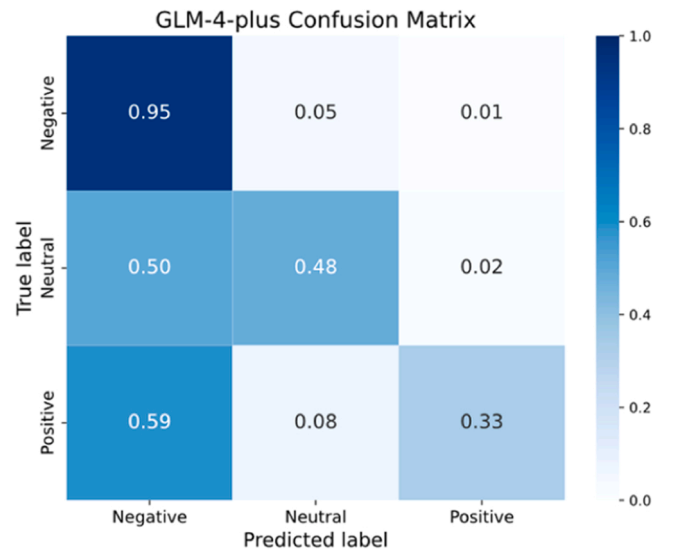


Fig. 7. Confusion Matrix of GLM-4-plus.

"Negative" categories. This indicates that while the model can grasp the general sentiment polarity, it struggles to discern whether the sentiment favours the euro or the dollar. Such findings highlight the necessity of employing a specialized classification framework tailored for sentiment analysis in forex-related news.

The classification results of the TF-IDF method further underscore its limitations in interpreting the semantics of exchange rate news, as depicted in Fig. 3. On one hand, the model demonstrates a notable tendency to misclassify many samples as "Neutral." On the other hand, it exhibits a pattern similar to the NoFT-FinBERT in terms of "confusion," evenly distributing predictions between "Positive" and "Negative." This is because frequency-based methods like TF-IDF fail to capture the nuanced semantic context where both "euro strengthening" and "dollar weakening" are positively correlated with the EUR/USD exchange rate.

Fig. 4–7 shows the confusion matrices for GPT-4, Claude-3.5-sonnet, Qwen-plus, and Glm-4-plus. While these large language models demonstrate relatively better performance compared to the NoFT-FinBERT, they still suffer from a significant number of misclassifications. Specifically, GPT-4, Claude-3.5-sonnet and Glm-4-plus tend to overclassify news as "Negative," Qwen-plus frequently

misclassifies news as "Neutral".

These findings highlight the necessity of fine-tuning models specifically for forex-related news. While the NoFT-FinBERT can only recognize sentiment polarity, and these most advanced LLMs exhibit some capacity to distinguish whether news is favorable for the euro or the dollar, they still produce considerable misclassifications. Thus, directly applying these models to exchange rate forecasting is inadequate without further domain-specific adaptation.

Another critical limitation of directly using above LLMs for sentiment analysis lies in the issue of potential data leakage. Our dataset spans from 2009 to 2023, with the objective of conducting "out-of-sample" yearly predictions from 2016 to 2023 to robustly demonstrate that sentiment indices enhance exchange rate forecasting accuracy. However, the training data for these LLMs evidently includes information from this time period, leading to the risk of future information leakage when utilizing them for classification.

4.2.2. Fine-tune for EUR/USD

In light of the two issues discussed in the previous section: (1) the distinct sentiment classification scheme required for forex-related news

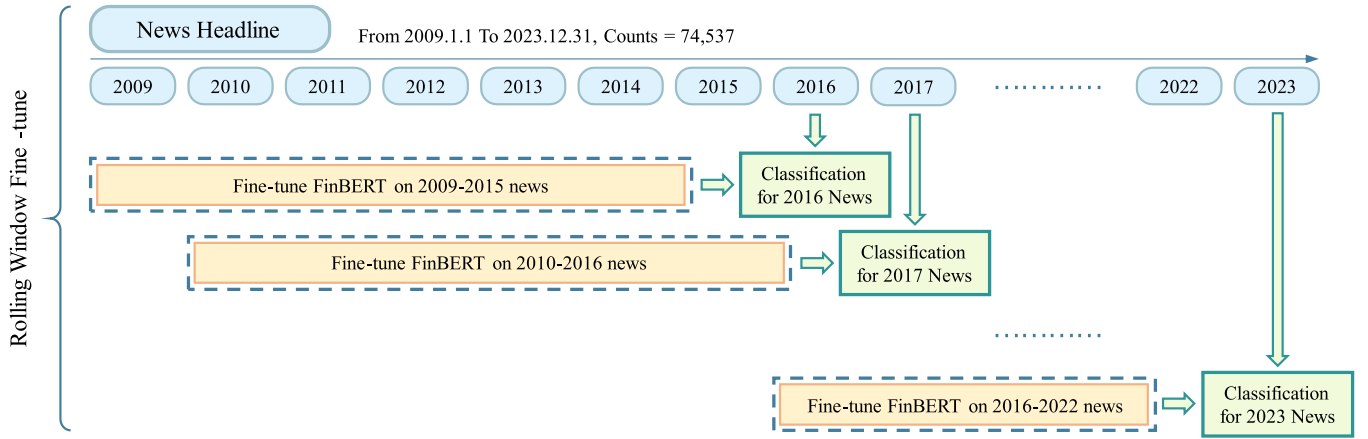


Fig. 8. Sentiment analysis workflow.

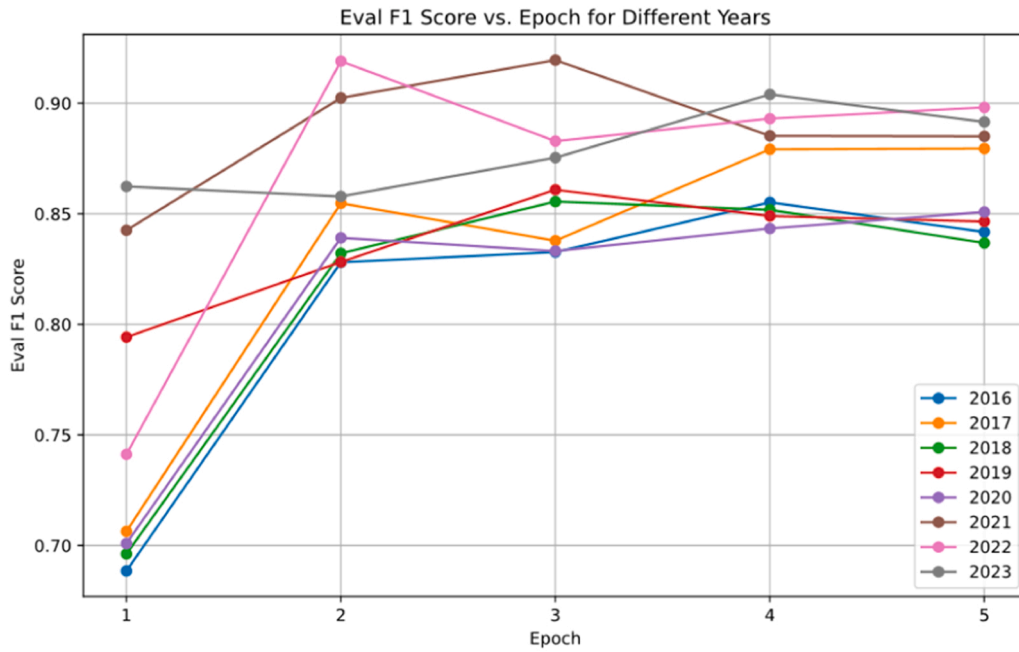


Fig. 9. Fine-tune evaluation score over Epochs.

and (2) the risk of information leakage when using proprietary LLM, we implemented a "rolling" fine-tuning approach to ensure the credibility of prediction results. Specifically, starting from 2009, the model was fine-tuned using seven years of data (2009–2015) to classify news for the eighth year (2016). This process was iteratively rolled forward until data from 2016 to 2023 were classified year by year, as illustrated in Fig. 8. As a result, we obtained eight versions of the model fine-tuned over different historical time windows, collectively referred to as FXFinBERT-EURUSD.

This approach not only addressed the aforementioned issues but also introduced a significant advantage: it enabled us to perform analyses over a long time span, thereby yielding convincing results that demonstrate the robustness of FXFinBERT across temporal and macroeconomic variations.

In conclusion, we developed a robust framework for annually updating FXFinBERT to handle the unique and dynamic challenges of forex-related sentiment analysis. This framework also establishes a solid foundation for future research in similar fields.

During each iteration, the model was trained for five epochs, with cross-entropy loss used as the loss function, as defined in Eq. (7). F1

Table 3

Performance metrics of the FXFinBERT-EURUSD.

Model	Eval Acc.	Eval F1.	Eval Loss.	Train FLOPs
FXFinBERT-EURUSD-2016	0.8435	0.8417	0.5566	1.19e+ 15
FXFinBERT-EURUSD-2017	0.8816	0.8795	0.4739	1.25e+ 15
FXFinBERT-EURUSD-2018	0.8387	0.8368	0.7732	1.32e+ 15
FXFinBERT-EURUSD-2019	0.8462	0.8465	0.7770	1.36e+ 15
FXFinBERT-EURUSD-2020	0.8509	0.8508	0.6197	1.40e+ 15
FXFinBERT-EURUSD-2021	0.8850	0.8850	0.4184	1.43e+ 15
FXFinBERT-EURUSD-2022	0.8986	0.8981	0.4376	1.46e+ 15
FXFinBERT-EURUSD-2023	0.8913	0.8915	0.4589	1.49e+ 15

score and Accuracy were employed as evaluation metrics to assess model performance. Fig. 9 clearly illustrates that the model's performance improved progressively with each epoch.

$$Loss = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^3 y_{ij} \log(\hat{y}_{ij}) \quad (18)$$

For FXFinBERT-EURUSD-2016, trained on data from 2009 to 2015, the F1 score increased from 0.6885 in the first epoch to 0.8417 by the

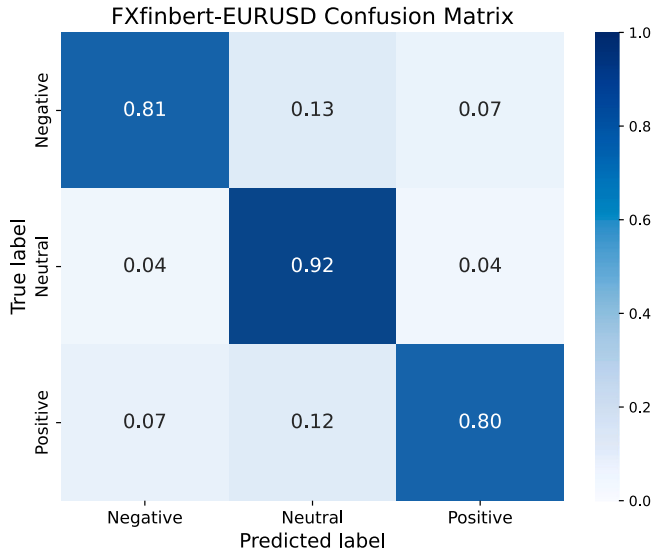


Fig. 10. Confusion matrix of FXFinBERT-EURUSD.

Table 4

Classification performance report of FXFinBERT-EURUSD.

Method	Macro P. ^{a)}	Macro R. ^{b)}	Macro F. ^{c)}	Acc. ^{d)}
FXFinBERT-EURUSD	84.86 %	84.34 %	84.60 %	84.33 %

fifth epoch. The best performance was observed with FXFinBERT-EURUSD-2022, fine-tuned on data from 2015 to 2021, achieving an F1 score of 0.8981 in the fifth epoch. Overall, the performance of each model consistently improved through fine-tuning. Table 3 summarizes the classification accuracy, F1 scores, and loss values for all models on the validation set, alongside the total floating-point operations (FLOPs) executed throughout the training process.

Through the rolling fine-tuning process, we ensured that the model performed classification and validation on out-of-sample data. Upon completing the year-by-year fine-tuning, we obtained the confusion matrix shown in Fig. 10, which integrates classification results from FXFinBERT-EURUSD-2016 to FXFinBERT-EURUSD-2023. These results demonstrate that fine-tuning significantly improved model performance, surpassing that of all proprietary large language models.

Table 4 presents the same metrics as shown in Table 3 from the previous section. Unlike the proprietary models, FXFinBERT-EURUSD

no longer exhibited a bias toward misclassifying any specific category. The "Positive" and "Negative" classes achieved classification accuracies of 80 % and 81 %, respectively. Moreover, the macro-averaged Accuracy, Recall, and F1 scores improved substantially, reaching 84.86 %, 84.34 %, and 84.60 %, respectively. These results robustly demonstrate that the fine-tuned FXFinBERT-EURUSD accurately classifies sentiment in forex-related news, making it suitable for the subsequent calculation of sentiment indices.

Next, we employed the Learning Interpretability Tool (LIT) developed by Google [58] to provide intuitive visual explanations of FXFinBERT-EURUSD. FXFinBERT-EURUSD-2023 was selected as a representative example for visualization. Two news articles correctly classified as "Positive" and two correctly classified as "Negative" were chosen, each published on different dates. For each sentence, LIT computed the importance ("salience" value) of each word in contributing to the model's "Positive" or "Negative" prediction based on back-propagation gradients. In Fig. 23, the intensity of each word's background color corresponds to its salience value, with darker shades indicating greater contribution. The results reveal that, for the EUR/USD currency pair, the model effectively identifies sentences containing information about a bearish USD or bullish EUR as "Positive," and conversely, sentences indicating a bullish USD or bearish EUR as "Negative."

4.2.3. Sentiment processing

After obtaining sentiment classification results for all years using the FXFinBERT models, we constructed daily sentiment indices for further analysis. For each trading day t , we computed the daily count and daily sentiment score for each sentiment category (positive, neutral, negative) as follows:

- **Daily Count ($C_{t,k}$)** for sentiment k :

$$C_{t,k} = \sum_{i \in N_t} 1(s_i = k), \# \quad (19)$$

where N_t is the set of news on day t , s_i is the sentiment label of news i , and $k \in \text{positive, neutral, negative}$.

- **Daily Sentiment Score ($S_{t,k}$)** for sentiment k :

$$S_{t,k} = \sum_{i \in N_t} w_i (\pm 1)(s_i = k), \# \quad (20)$$

where w_i is the predictive score of news i , ± 1 means a positive new will multiply by 1 and a negative new will multiply by -1 .

Missing sentiment data was treated as zero. Finally, sentiment data

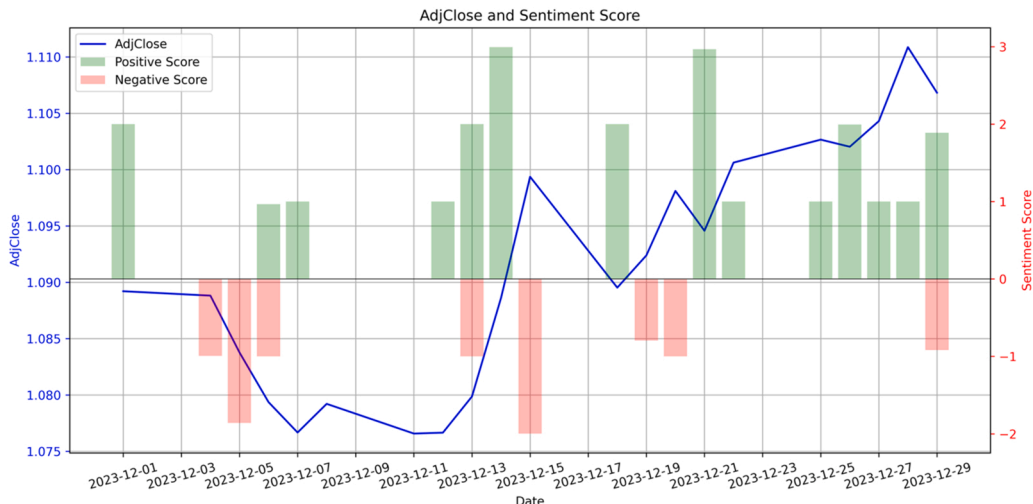


Fig. 11. EUR/USD AdjClose and daily sentiment score.

Category	Abbrev.	Correlation Test	Granger Test	LGBM Importance	XGB Importance	RF Importance	RFECV Rank	Boruta Rank
Senti-Score Index	$ScSS_t$	✓	0.00	0.05	0.10	0.10	1	1
	$ScPS_t$	✓	0.00	0.05	0.08	0.20	1	1
	$ScNS_t$	✓	0.00	0.05	0.11	0.24	1	1
	$AbSS_t$	✗						
Senti-Change Index	$AdSS_t$	✓	0.00	0.03	0.06	0.16	6	1
	SC_t	✓	0.72	0.01	0.03	0.01	15	11
Senti-Continuity Index	SP_t	✓	0.00	0.01	0.02	0.00	26	20
	$ScSS_{t-1}, ScSS_{t-2}, ScSS_{t-3}$	✓ ✓ ✓	0.00 0.00 0.00	0.02 0.01 0.01	0.03 0.02 0.02	0.01 0.01 0.01	5 22 21	12 5 7
Lag\Diff\Avg Index	$SC_{t-1}, SC_{t-2}, SC_{t-3}$	✓ ✓ ✓	0.30 0.45 0.59	0.01 0.02 0.02	0.03 0.03 0.04	0.01 0.01 0.03	9 4 8	7 3 1
	$SP_{t-1}, SP_{t-2}, SP_{t-3}$	✗ ✗ ✓	0.00	0.01	0.03	0.01	17	15
	$\Delta ScSS_t, \Delta^2 ScSS_t, \Delta^3 ScSS_t$	✓ ✓ ✓	0.00 0.00 0.00	0.02 0.01 0.01	0.03 0.02 0.02	0.01 0.01 0.01	11 16 18	4 2 5
	$\Delta SC_t, \Delta^2 SC_t, \Delta^3 SC_t$	✓ ✓ ✓	0.46 0.24 0.57	0.01 0.01 0.01	0.02 0.02 0.02	0.01 0.01 0.01	23 14 24	9 9 14
	$\Delta SP_t, \Delta^2 SP_t, \Delta^3 SP_t$	✓ ✓ ✓	0.00 0.00 0.00	0.01 0.01 0.01	0.03 0.02 0.02	0.02 0.01 0.01	10 19 13	1 16 18
	$MA_1(ScSS_t), MA_2(ScSS_t), MA_3(ScSS_t)$	✓ ✓ ✓	0.00 0.00 0.00	0.01 0.01 0.01	0.01 0.02 0.01	0.00 0.01 0.01	20 25 27	21 13 16
	$MA_1(SC_t), MA_2(SC_t), MA_3(SC_t)$	✓ ✓ ✓	0.43 0.47 0.76	0.02 0.03 0.03	0.04 0.05 0.05	0.02 0.04 0.04	12 3 2	1 1 1
	$MA_1(SP_t), MA_2(SP_t), MA_3(SP_t)$	✗ ✗ ✓	0.00	0.01	0.02	0.01	7	18

Fig. 12. News data selection heatmap.

was merged with FX data into a single dataset for subsequent modelling.

Subsequently, we created a dual-axis plot focusing on the period from December 1, 2023, to December 29, 2023, as shown in Fig. 11. The plot displays the daily closing prices (Close) of the EUR/USD currency pair as a line graph, while the daily positive and negative sentiment scores are represented by green and red bar graphs, respectively. The chart clearly demonstrates that news sentiment has predictive power for changes in exchange rates.

4.2.4. Sentiment index construction and selection

To deepen our understanding of investor sentiment and its influence on exchange rate dynamics, we constructed a sentiment index pool based on three key dimensions: intensity, change, and persistence. These dynamic features directly reflect the foundational view introduced in the Introduction—that price volatility, as captured by models such as GARCH, evolves along these same three axes. Guided by this three-dimensional perspective, we developed a set of sentiment indices grouped into three categories: the Senti-Score Index (measuring absolute sentiment intensity), the Senti-Change Index (capturing directional shifts in sentiment), and the Senti-Continuity Index (quantifying sentiment persistence over time). First, by summing the intraday news sentiment scores, we obtained the Sentiment Score (SS_t) of a day.

$$SS_t = S_{t,positive} + S_{t,negative} \# \quad (21)$$

where t represents the t -th day with news records.

Since the number of daily news texts was clearly unbalanced over time, to avoid overfitting due to data distribution issues, we standardized SS_t to obtain the Scaled Sentiment Score ($ScSS_t$).

$$ScSS_t = \frac{S_{t,positive} + S_{t,negative}}{C_{t,positive} + C_{t,negative} + 1} \# \quad (22)$$

Similarly, we derived Scaled Positive Score ($ScPS_t$) and Scaled Negative Score ($ScNS_t$) to distinguish positive and negative sentiment intensity; Absolute Sentiment Scale ($AbSS_t$) was obtained by normalizing SS_t to 0–1, used to measure absolute sentiment intensity.

The effectiveness of sentiment indices may have been diminished when there was a strong expectation for a certain trend. We believed that if a significant negative news item appeared during a relatively stable period of Euro appreciation, it could have potentially triggered a substantial reversal effect. Therefore, we needed to apply a certain degree of amplification or diminution to the original news index. We constructed Adjusted Sentiment Scale ($AdSS_t$), adjusted by the 15-day EUR/USD exchange rate closing price change rate, to account for sentiment expectations.

$$AdSS_t = \frac{AbSS_t \cdot Close_{t-15}}{Close_t} \# \quad (23)$$

Based on Senti-Score Index, we provided simple examples of Senti-Change Index and Senti-Continuity Index, two types of indices measuring Sentiment Changes (SC_t) and Sentiment Persistence (SP_t), respectively.

$$SC_t = (ScPS_t - ScNS_t) - (ScPS_{t-1} - ScNS_{t-1}) \# \quad (24)$$

$$SP_t = \sum_{i=\max(0,t-7)}^{t-1} e^{-\frac{t-i}{7}} \cdot ScSS_i + ScSS_t \# \quad (25)$$

Finally, we performed $\text{lag}(t-1, t-2, t-3)$, $\text{diff}(\Delta, \Delta^2, \Delta^3)$, and $\text{avg}(MA_1, MA_2, MA_3)$ operations on above indices to construct the complete index pool. The results of each index selection method are

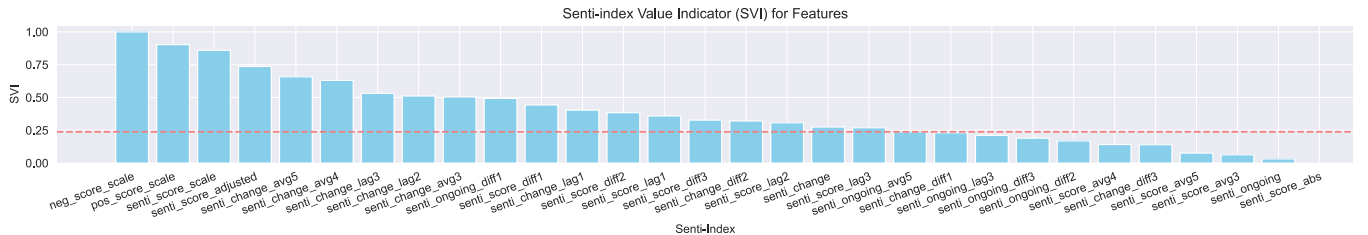


Fig. 13. SVI bar chart.

Table 5
DDML regression results 1.

	$S_{t,negative}$ (1)	$S_{t,positive}$ (2)	$S_{t,negative}$ (3)	$S_{t,positive}$ (4)
$\Delta_{t,1}$	2.604***	1.567***	7.428***	7.657***
$\Delta_{t,3}$	2.367***	1.385**	8.628***	8.587***
$\Delta_{t,7}$	1.999***	0.087	7.274***	7.540***
$\Delta_{t,30}$	4.032**	-0.691	4.431***	5.723***
$\Delta_{t,90}$	7.539***	-3.436	8.118***	2.370
TI FE	T	T	T	T
MI FE	F	F	T	T
Observations	5476	5476	5476	5476

Notes: Significance level: * 0.10, ** 0.05, *** 0.01. This applies to subsequent tables.

Table 6
DDML regression results 2.

	$ScPS_t$ (5)	$ScNS_t$ (6)	$AdSS_t$ (7)	SP_t (8)
$\Delta_{t,1}$	71.331***	71.132***	92.762***	12.054***
$\Delta_{t,3}$	79.630***	81.428***	101.581***	11.225***
$\Delta_{t,7}$	67.330***	73.278***	89.564***	6.262***
$\Delta_{t,30}$	59.164***	56.139***	76.225***	-2.776
$\Delta_{t,90}$	76.749***	62.747***	92.195***	15.524**
TI FE	T	T	T	T
MI FE	T	T	T	T
Observations	5476	5476	5476	5476

summarized in Fig. 12, where we have drawn a heatmap based on their importance. The darker the color, the more important the index is. We dropped the indices which SVI is in the bottom 25 %. The index selection results are shown in Fig. 13.

4.3. In-sample analysis

4.3.1. Fixed effects and double debiased machine learning (DDML) results

We first present the average treatment effects (ATE) of the untreated Daily Sentiment Scores, $S_{t,negative}$ and $S_{t,positive}$, on future exchange rate changes over the horizons from 1 to 90 days ahead, while controlling for fixed effects from Technical Indicators (TI FE) and Macroeconomic Indicators (MI FE) in Table 5. The average treatment effects are expressed in basis points (bp), where one basis point is equivalent to a 0.01 % change in the exchange rate. This allows us to quantify the impact of market sentiment on exchange rate movements in a standardized manner.

As defined in Eq. (20), the value of $S_{t,negative}$ is negative, and $S_{t,positive}$ is positive. Consequently, a positive ATE indicates that negative market sentiment drives the exchange rate downward, while positive sentiment drives the exchange rate upward, consistent with our definitions of Negative and Positive sentiment. Furthermore, we observe that in models (1) and (3), the ATE of $S_{t,negative}$ either increases or remains stable as the prediction horizon lengthens. In contrast, models (2) and (4) show a declining ATE of $S_{t,positive}$ as the forecast horizon extends, with the ATE of $S_{t,positive}$ being consistently lower than that of $S_{t,negative}$ for same forecast period.

Furthermore, in Table 6, we report the ATEs of the processed sentiment indices $ScPS_t$, $ScNS_t$, $AdSS_t$ and SP_t on future exchange rate changes from 1 to 90 days ahead, while controlling for fixed effects from Technical Indicators (TI FE) and Macroeconomic Indicators (MI FE). The results show that, compared to the raw Daily Sentiment Scores, the sentiment indices generally have higher ATE on exchange rate changes across all forecast periods. This indicates that transforming raw news sentiment into sentiment indices leads to a better representation of market sentiment. Among these indices, the Adjusted Sentiment Scale $AdSS_t$ shows an ATE of approximately 100 bps on 3-day exchange rate changes.

Table 7
DDML regression results 3.

	$ScNS_t^{Qwen}$ (9)	$AdSS_t^{Qwen}$ (10)	$ScNS_t^{TF-IDF}$ (11)	$AdSS_t^{TF-IDF}$ (12)
$\Delta_{t,1}$	56.444***	81.135***	11.387***	27.430***
$\Delta_{t,3}$	60.340***	84.651***	13.706***	33.743***
$\Delta_{t,7}$	52.036***	71.108***	14.899**	33.192***
$\Delta_{t,30}$	42.955***	63.431***	23.169*	47.498**
$\Delta_{t,90}$	38.357*	58.588**	21.576	51.750
TI FE	T	T	T	T
MI FE	T	T	T	T
Observations	5476	5476	5476	5476

Lastly, in Table 7, we report the DDML regression results for the sentiment indices constructed using Qwen-plus and the TF-IDF method. Using the Scaled Negative Score ($ScNS_t$) and Adjusted Sentiment Scale ($AdSS_t$) as examples, we find that the ATEs of both methods remain significant over shorter horizons but are noticeably lower than the results from models (6) and (7) in Table 6, indicating that sentiment information influencing exchange rates was not well captured by these two methods. Nonetheless, the Qwen-plus sentiment indices still show higher ATEs than those derived from TF-IDF, reinforcing our earlier analysis that LLMs possess far superior sentiment understanding capabilities compared to traditional methods. However, due to the unique nature of forex-related news sentiment, directly applying LLMs is less effective than fine-tuning FXFinBERT for specific exchange rate tasks.

4.3.2. Theoretical mechanisms and empirical evidence

The findings presented in Section 4.3.1 align with and provide empirical validation for the theoretical mechanisms proposed in prior literature regarding the influence of news sentiment on exchange rates [24,59,60]. Economic theory suggests that negative sentiment tends to trigger stronger and more prolonged reactions in financial markets due to risk aversion and flight-to-safety behaviors [59,61–63]. Our results, as shown in Table 5, confirm this phenomenon: the Average Treatment Effects (ATEs) of negative sentiment scores ($S_{t,negative}$) are consistently larger and more persistent across longer forecast horizons compared to positive sentiment scores ($S_{t,positive}$). This asymmetry highlights the market's heightened sensitivity to adverse news, which may reflect investors' precautionary behavior when faced with uncertainty.

Furthermore, the enhanced predictive power of processed sentiment indices, as reported in Tables 6 and 7, supports the view that sentiment indices more effectively aggregate relevant information from news, compared to raw sentiment scores. The Adjusted Sentiment Scale ($AdSS_t$) demonstrates particularly strong predictive effects, with ATEs reaching up to 100 basis points over shorter horizons. This result reinforces the theoretical argument that refined sentiment measures—capturing nuanced information through advanced processing methods—are critical for modeling sentiment-driven exchange rate movements [22,64–66].

4.4. Out-of-sample prediction

4.4.1. Performance evaluation of models with sentiment indices

As previously mentioned, to better capture the impact of news indices, we set the target of our model predictions as $\Delta_{t,1} = Close_{t+1} - Close_t$. We compared various models including neural network (LSTM), ensemble learning models (LightGBM, RF, GBDT), and machine learning models (KNN, ENR, SVR) to assess the effectiveness of news indices. The predicted results are added back to the exchange rate of the current day to derive the forecasted exchange rate for the following day. The experiments employed MAE, MSE, MSLE, MAPE, and LogCosh metrics to evaluate and compare the forecasting accuracy of these seven models with and without news indices. Additionally, we constructed ARIMA model with automatic order selection and GARCH (1,1) model as baselines, which underwent rolling training on the data

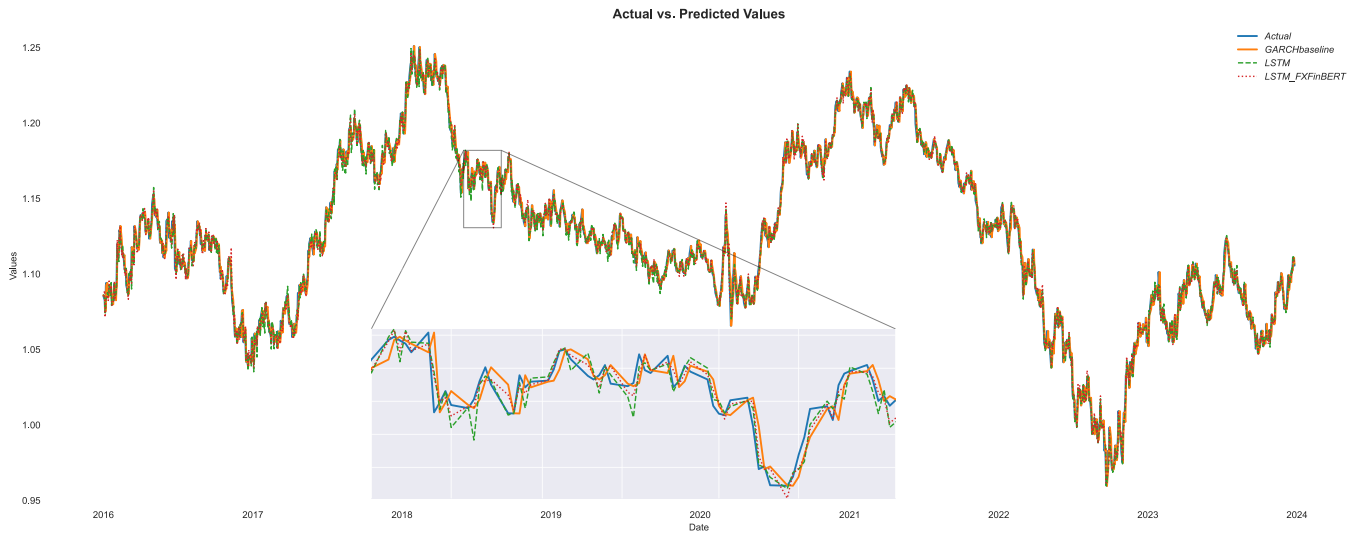


Fig. 14. Comparison of GARCH and LSTM Models With or Without Sentiment Indices.

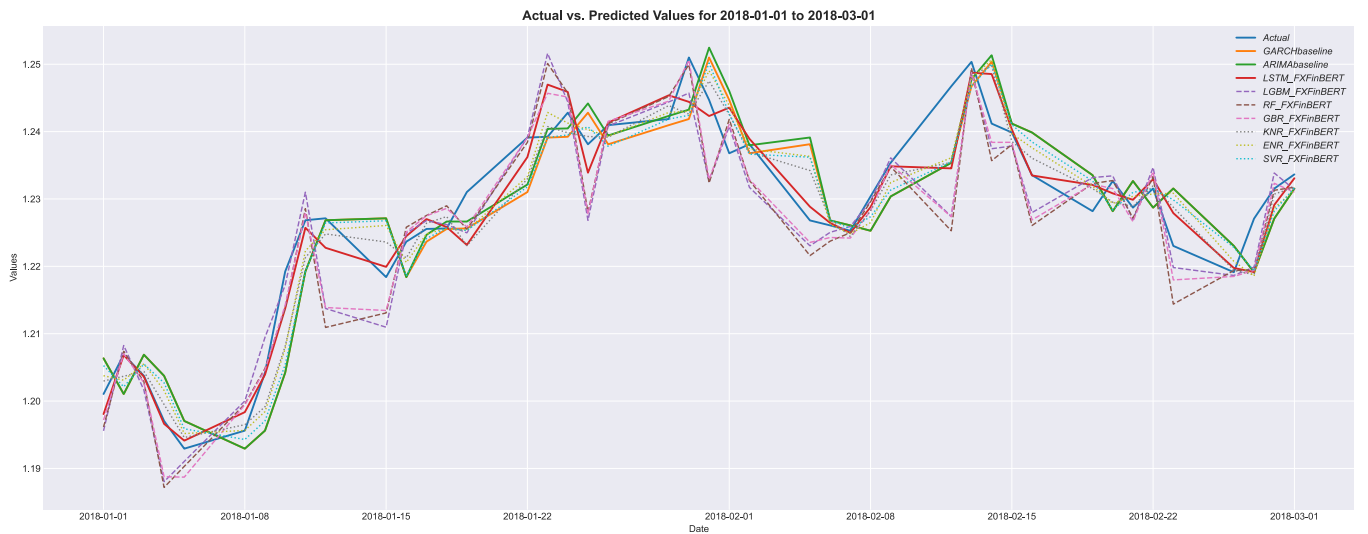


Fig. 15. Comparison of All Models with Sentiment Indices.

and performed one-step ahead predictions for comparison with other models.

Initially, the experiments involved rolling training with the seven models using data from 2009 to 2023. Each training session utilized a seven-year data span. For instance, in the first iteration, data from 2009 to 2015 were used for training, followed by rolling one-step ahead predictions on the data from 2016, akin to the rolling fine-tuning process mentioned earlier for FinBERT. During the training of each model, both the training and test sets underwent separate standard normalization.

Subsequently, the experiments conducted rolling forecasts for each model under including or excluding these sentiment indices calculated by the FXFinBERT. This allowed us to obtain a series of predictions and corresponding metrics, forming the basis for our analysis of the effect of sentiment indices.

Fig. 14 demonstrates the improvement in predictive accuracy of the LSTM neural network model with the addition of sentiment indices. For the highly volatile EUR/USD exchange rate series, the benchmark GARCH model tends to use the previous day's actual value as the prediction for the next day. The LSTM model we constructed without sentiment indices can capture temporal features through intra-day and inter-day technical indicators, forming relatively effective predictions.

In comparison, the LSTM model with sentiment indices provides more stable predictions for time series fluctuations. It can be observed in the figure that at some turning points of exchange rate fluctuations, the LSTM model without sentiment indices often amplifies the volatility effect (green dashed line), while the LSTM model with sentiment indices can adjust the volatility to an appropriate level (red dashed line).

We further compared the learning ability of different models for the sentiment indices. To present our conclusions more clearly, we plotted the predicted value curves of each model with sentiment indices between January 1, 2018, and March 1, 2018, as shown in Fig. 15. In this figure, different types of models exhibit obvious similarities. Three machine learning models: KNR, ENR, and SVR show a phenomenon similar to ARIMA and GARCH, using the previous day's actual value as the prediction. The prediction results of the three ensemble learning models: LGBM, RFR, and GBR have a very high degree of overlap and tend to amplify the volatility of one-step predictions. The LSTM model with sentiment indices demonstrates good predictive ability at the turning points of exchange rate fluctuations—it can predict the situation when a trend change occurs—this comes from LSTM's better temporal feature learning ability and the trend tendency reflected by the sentiment indices.

Table 8

Comparison results of model evaluation indicators.

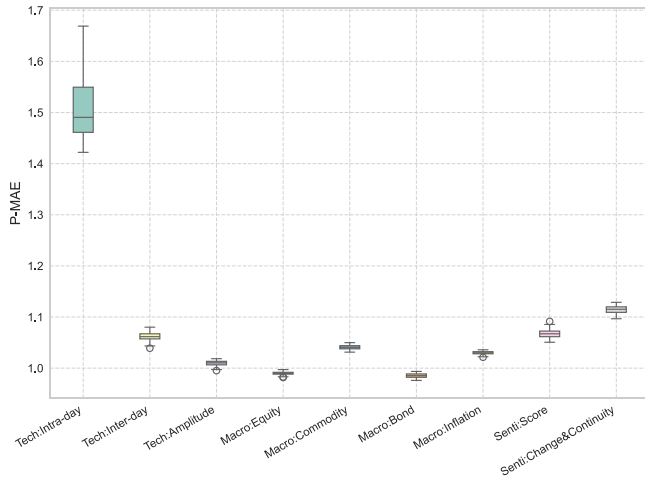
Model	MAE	IR	MSE	IR	MSLE	IR	MAPE	IR	LogCosh	IR	Rank
LSTM^{SI}	2.198	7.5 %	0.962	11.4 %	2.143	11.7 %	1.961	7.6 %	4.810	11.4 %	1
LSTM	2.376		1.086		2.426		2.122		5.428		2
LGBR^{SI}	2.623	5.0 %	1.309	10.9 %	2.912	10.8 %	2.340	5.0 %	6.546	10.9 %	3
LGBR	2.760		1.469		3.264		2.463		7.344		5
RFR^{SI}	2.663	5.3 %	1.424	9.0 %	3.164	9.1 %	2.376	5.3 %	7.120	9.0 %	4
RFR	2.811		1.565		3.479		2.509		7.827		7
GBR^{SI}	2.748	5.7 %	1.491	10.6 %	3.320	10.5 %	2.454	5.6 %	7.455	10.6 %	6
GBR	2.913		1.668		3.707		2.600		8.342		8
KNR^{SI}	3.061	9.1 %	1.699	13.3 %	3.809	13.2 %	2.739	9.1 %	8.495	13.3 %	9
KNR	3.366		1.959		4.391		3.013		9.797		11
ENR^{SI}	3.132	9.6 %	1.757	16.2 %	3.933	16.2 %	2.803	9.6 %	8.785	16.2 %	10
ENR	3.465		2.097		4.693		3.100		10.485		13
SVR^{SI}	3.317	10.2 %	2.034	17.7 %	4.563	17.1 %	2.968	10.0 %	10.170	17.7 %	12
SVR	3.693		2.471		5.504		3.297		12.354		14
GARCH	3.935		2.676		5.998		3.521		13.380		15
ARIMA	4.034		2.838		6.359		3.610		14.187		16

Notes: ^{SI} means model with sentiment indices. The metric values are presented in scientific notation: MAE (e-03), MSE (e-05), MSLE (e-06), MAPE (e-01), and LogCosh (e-06). 'IR' is calculated as: $IR_{model} = \frac{Metric_{model} - Metric_{model}^{SI}}{Metric_{model}}$.

Table 9

DM TEst Results: Model Performance with and without Sentiment Index.

Model	DM
LSTM	3.6012***
SVR	4.4288***
RFR	3.0803***
GBR	2.9377***
LGBM	4.1784***
ENR	22.3932***
KNR	8.8272***

**Fig. 16.** Permutation Importance.

We recorded the value of each loss function for each model in Table 8, where IR represents the improvement ratio of predictive accuracy for each model after incorporating sentiment indices. Here, we can also observe the similarity in predictive performance across different types of models. LSTM, which already had the best predictive performance, becomes even better. Ensemble learning models all show an improvement of around 10 % after incorporating sentiment indices (taking MSE as an example). Finally, we obtained the final ranking of model predictive performance, which is shown in the "Rank" column.

4.4.2. Further analysis of models with sentiment indices

We conducted a DM test to compare the prediction results of each model before and after adding the news sentiment index pools. The results showed consistent improvements in predictive performance across all models when sentiment data was included. The DM test statistics showed significant increases, with p-values well below conventional thresholds, indicating that the improvements were statistically meaningful rather than random. This highlights the value of incorporating sentiment indices into predictive models, as they capture contextual information that traditional numerical features might miss.

Next, we used MAE as the loss function to evaluate the LSTM model based on permutation importance and compared the importance of the sentiment indices with technical indicators and macroeconomics indicators, as shown in Fig. 16. The results indicated that intra-day technical indicators hold the highest importance. Sentiment indices followed in importance, and once sentiment score indices or sentiment change&continuity indices were permuted, the model's error increased by an average of 8 % or 10 %. To further validate these findings, we conducted comprehensive ablation studies by systematically removing each feature category 50 times for model training. As demonstrated in Table 13, the removal of intra-day technical indicators caused the most dramatic performance degradation, confirming their dominant role in the prediction system. Notably, sentiment features showed substantial impact - ablating change&continuity indices increased MAE by 11.2 % ($p < 0.001$), while score indices removal led to 7.3 % performance drop ($p < 0.001$). Complete results for MSE, MSLE, MAPE and LogCosh metrics are also provided in Appendix B.

4.4.3. Comparison of sentiment analysis approaches for exchange rate forecasting

By incorporating sentiment indices derived from different methods into the LSTM model, we compare the differences between various exchange rate news sentiment analysis approaches. The results are presented in Fig. 17. Across five prediction accuracy metrics, the sentiment index constructed using the FXFinBERT model significantly enhanced the predictive accuracy of the LSTM model. In contrast, the inclusion of the sentiment index based on the Qwen-plus model only improved prediction accuracy slightly in terms of MAE and MAPE, while the sentiment index constructed using the TF-IDF method led to a deterioration in accuracy across all metrics. Based on the earlier discussion in Section 4.2.1, the observed results align with the argument that fine-tuning sentiment analysis models is crucial for enhancing predictive accuracy. The superior performance of the FXFinBERT sentiment index underscores the importance of a tailored sentiment classification system

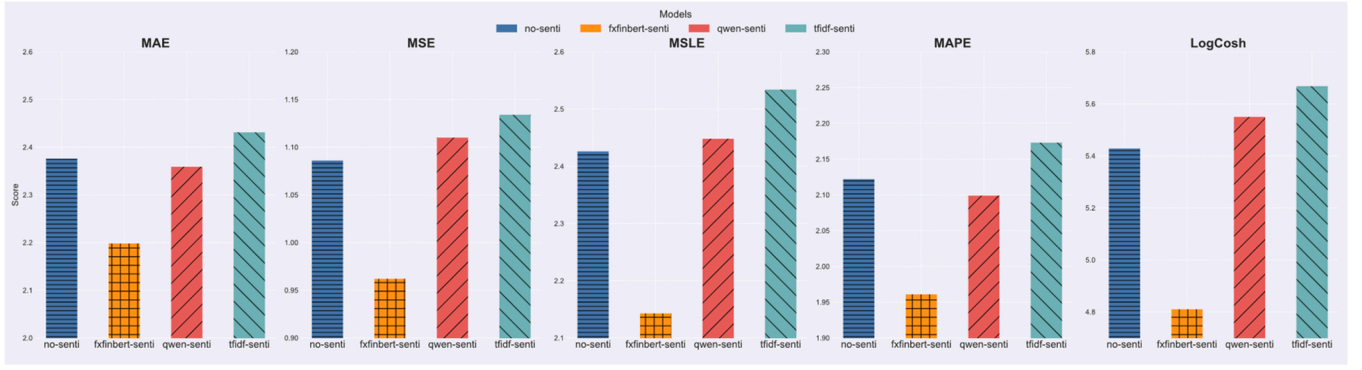


Fig. 17. Comparison of LSTM with Different Sentiment Analysis Approaches Notes: In the figure, "no-senti", "fxfinbert-senti", "qwen-senti", and "tfidf-senti" respectively represent predictions without sentiment indices, with sentiment indices from FXFinBERT, from Qwen-plus, and from TF-IDF method. The metrics are presented using the same scientific notation as in Table 8, the lower metric means better performance.

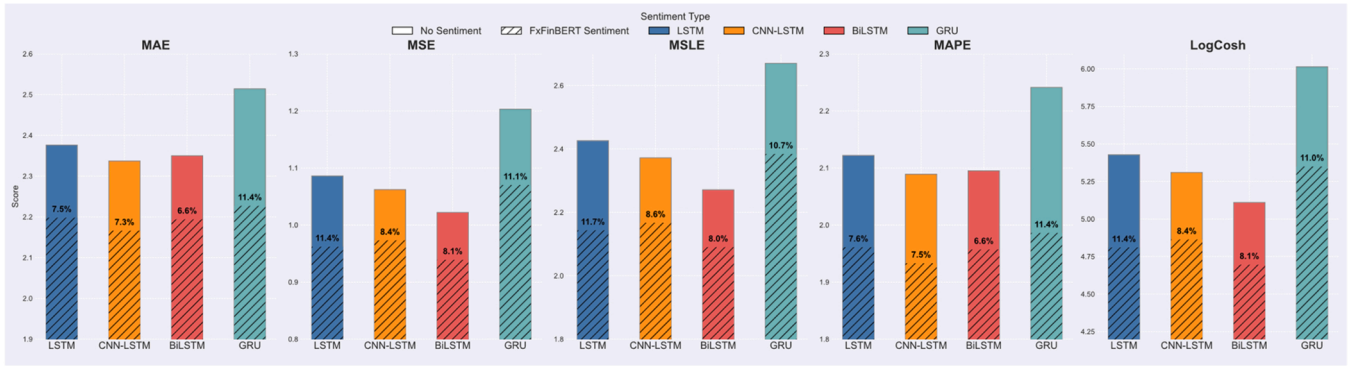


Fig. 18. Comparison of LSTM Family with and without Sentiment.

for FX-related news, different from other domains, where nuanced sentiment interpretation plays a pivotal role in forecasting accuracy.

4.4.4. Comparison with the Family of LSTM models and interval prediction

In this section, we examine whether incorporating sentiment indices into the LSTM model family and its various variants can robustly enhance predictive performance. The models compared include CNN-LSTM, BiLSTM, and GRU. The results, illustrated in Fig. 18, reveal the comparative performance. Different colors represent distinct models, and left-slanted lines indicate the performance achieved after integrating sentiment indices. The percentage increase in predictive performance is annotated on the bar charts. All predictive models demonstrate noticeable improvements upon incorporating sentiment factors, supporting the consistency and robustness of our conclusion. This further demonstrates the transferability of our approach. Combined with the results in Table 8, we can conclude that the sentiment index constructed by FXFinBERT significantly enhances the performance of various multivariate forecasting models.

We further extend point predictions to interval predictions to investigate the impact of uncertainty on forecast fluctuations, which is particularly significant for exchange rate prediction. Specifically, our model provides one-step forecasts for exchange rate movements, interpreted as expectations. The probability of the true value falling within the predicted lower and upper bounds is expressed as $P(L \leq \Delta_t \leq U) = P(L \leq \Delta_t \leq U | E(\Delta_t) = \widehat{\Delta}_t) \times P(E(\Delta_t) = \widehat{\Delta}_t)$. By assuming that the forecasted values share the same distribution as historical observations and that the variance of forecasts can be approximated using the variance S^2 from historical windows, the expected value $E(\Delta_t)$ and variance $Var(\Delta_t)$ are defined as: $\int \Delta_t f(\Delta_t | \theta) d\Delta_t = \widehat{\Delta}_t$ and $\int (\Delta_t - E(\Delta_t))^2 f(\Delta_t | \theta) d\Delta_t = S^2$. Consequently, at a significance level of

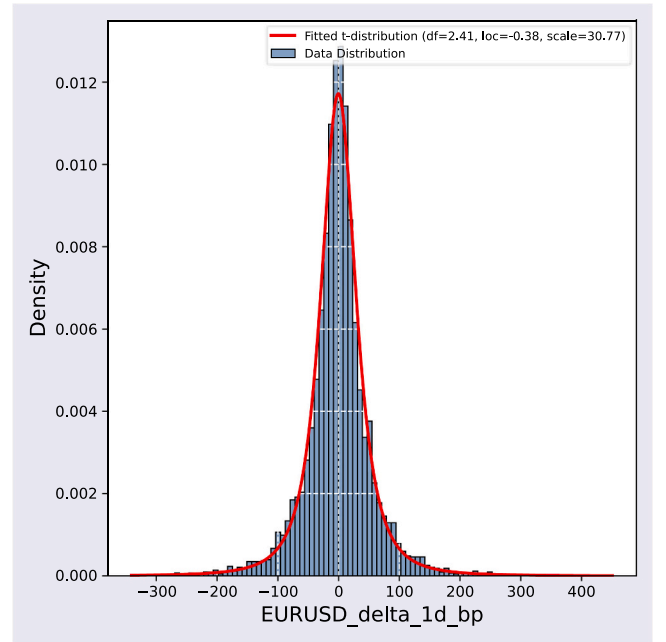


Fig. 19. Distribution of EUR/USD price change.

2α , the upper and lower bounds of the prediction interval $[L, U]$ satisfy: $P(L \leq \Delta_t \leq U) = 1 - 2\alpha$. We next establish the distributional characteristics of the true values.

First, we perform a series of normality tests on historical exchange

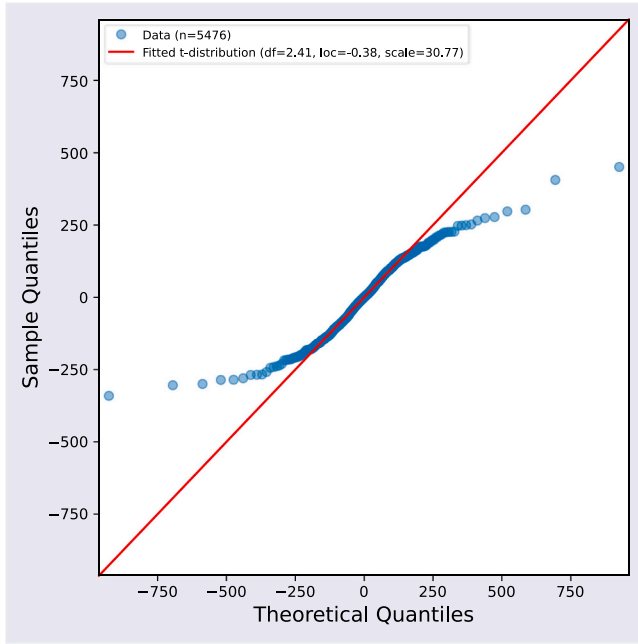


Fig. 20. Q-Q plot of EUR/USD price change.

rate changes ($\Delta_{t,1}$) using the Shapiro-Wilk test, Kolmogorov-Smirnov test, Anderson-Darling test, and Jarque-Bera test. As all tests reject the null hypothesis of normality and the data exhibit higher kurtosis, we fit the distribution of exchange rate changes using Student's t -distribution. The goodness-of-fit is evaluated via the Kolmogorov-Smirnov test, which at the 1 % significance level fails to reject the t -distribution assumption. The histogram of price changes overlaid with the fitted t -distribution density curve is presented in Fig. 19, while the Q-Q plot in Fig. 20 demonstrates the alignment of data with the fitted t -distribution. Here, the unit of price change is measured in basis points.

We employ a LSTM model enhanced with the FXFinBERT sentiment indices to predict daily EUR/USD exchange rate changes. Using the t -distribution derived above, we calculate prediction intervals, which are then added to the current day's price to obtain the interval forecast for the next day's price. In Fig. 21, the gray-shaded area represents the prediction interval for EUR/USD price at a 0.4 significance level, while the blue line indicates actual prices. The chart shows that the prediction

closely follows actual price movements, with almost true prices falling within the interval, indicating the model's capability for effective interval forecasting.

5. Conclusion

This study contributes to the literature on exchange rate forecasting by proposing a novel sentiment analysis model, FXFinBERT, tailored specifically for the EUR/USD currency pair. This approach addresses the shortcomings in existing research regarding the unique classification of sentiment in forex-related news as we mentioned earlier. Empirical results demonstrate that FXFinBERT outperforms traditional sentiment analysis tools and state-of-the-art large language models (LLMs) in capturing the nuanced sentiment of forex-related news, achieving an out-of-sample accuracy of 84.33 %. The constructed sentiment indices, validated using the doubly debiased machine learning (DDML) approach, exhibit significant predictive power for exchange rate movements across multiple time horizons. For instance, using $AdSS_t$ as a sentiment factor, the average treatment effects on 1-, 3-, 7-, 30-, and 90-day exchange rate changes are 92.762***, 101.581***, 89.564***, 76.225***, and 92.195***, respectively. These effects surpass those of original sentiment classifications, Qwen-plus-based sentiment factors, and TF-IDF-based sentiment factors.

Furthermore, the inclusion of refined sentiment indices improves the predictive performance of various forecasting models. For example, incorporating these indices into the LSTM model reduces mean absolute error (MAE) by 7.5 % and mean squared error (MSE) by 11.4 %. Robustness tests, including the DM test, feature importance rankings, comparisons across sentiment analysis methods, and evaluations of different LSTM variants, further validate these findings. Permutation Importance and Ablation Analysis reveal the relative importance of different categories of predictors. Specifically, when sentiment indices are permuted during the prediction phase or removed during the training phase, the model's MAE increases by 10 % and 11 %, respectively. Notably, the integration of sentiment indices enhances not only the accuracy of point forecasts but also the robustness of interval predictions, as evidenced by tighter prediction intervals.

Theoretically, our findings highlight the critical role of refined sentiment indices in capturing the complex dynamics of exchange rate movements. The observed asymmetry in the impact of positive and negative sentiment, where negative sentiment exerts a stronger and more persistent influence, aligns with economic theories on risk aversion and market reactions to adverse news.

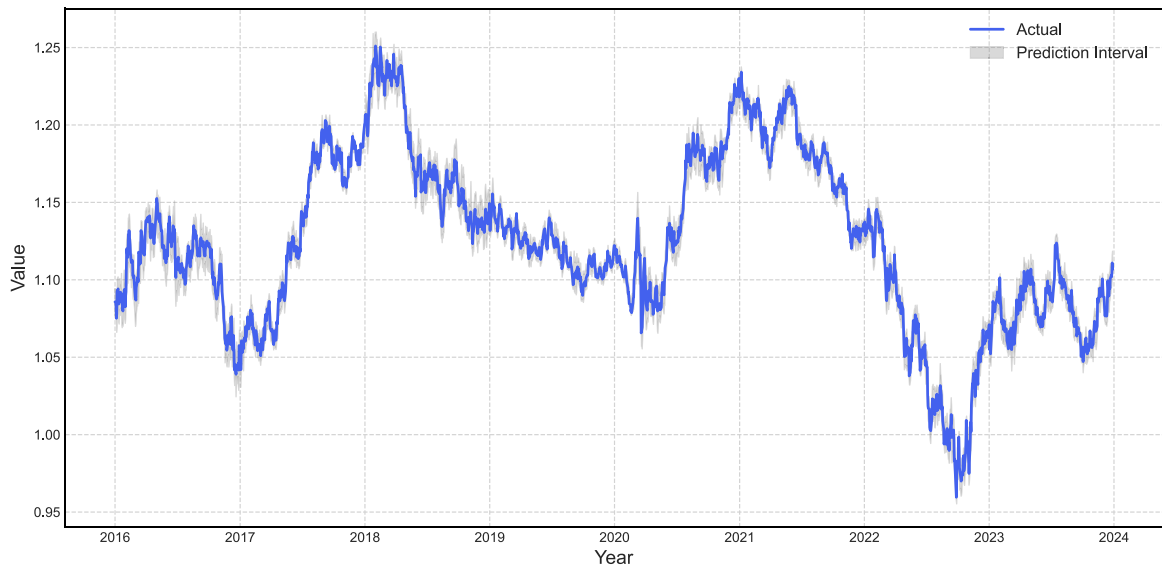


Fig. 21. The prediction intervals of EUR/USD price.

While this study provides robust empirical evidence and practical insights, several limitations remain, offering opportunities for future research. First, leveraging advanced large language model techniques, such as multi-agent systems or retrieval-augmented generation could potentially enhance classification performance [67]. Second, incorporating additional external data sources, such as social media or financial terminal information feeds, could further enhance the predictive power of sentiment indices, while such studies are widely applied in stock market analysis [68,69]. Third, future work could explore the transferability of FXFinBERT models: applying the model fine-tuned on EUR/USD to other currency pairs may yield promising performance, facilitating broader cross-market sentiment analysis.

In summary, this study makes dual contributions to the domain of sentiment-aware exchange rate forecasting. Methodologically, we propose FXFinBERT, a computationally efficient model (training FLOPs $\approx 1.3e+15$), specifically designed for prediction in highly volatile forex markets through an rolling training mechanism that continuously adapts to market semantics evolution. Empirically, we establish a dynamic sentiment features framework, revealing an asymmetric impact where negative sentiment exerts stronger influence on currency movements than positive sentiment - a finding robustly aligned with prospect theory's loss aversion principle. These advancements not only bridge the

gap between behavioral finance and quantitative forecasting but also provide a scalable architecture for real-time market sentiment analysis.

CRediT authorship contribution statement

Yuping Song: Writing – review & editing, Supervision, Project administration. **Guoqin Gu:** Writing – original draft, Software, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

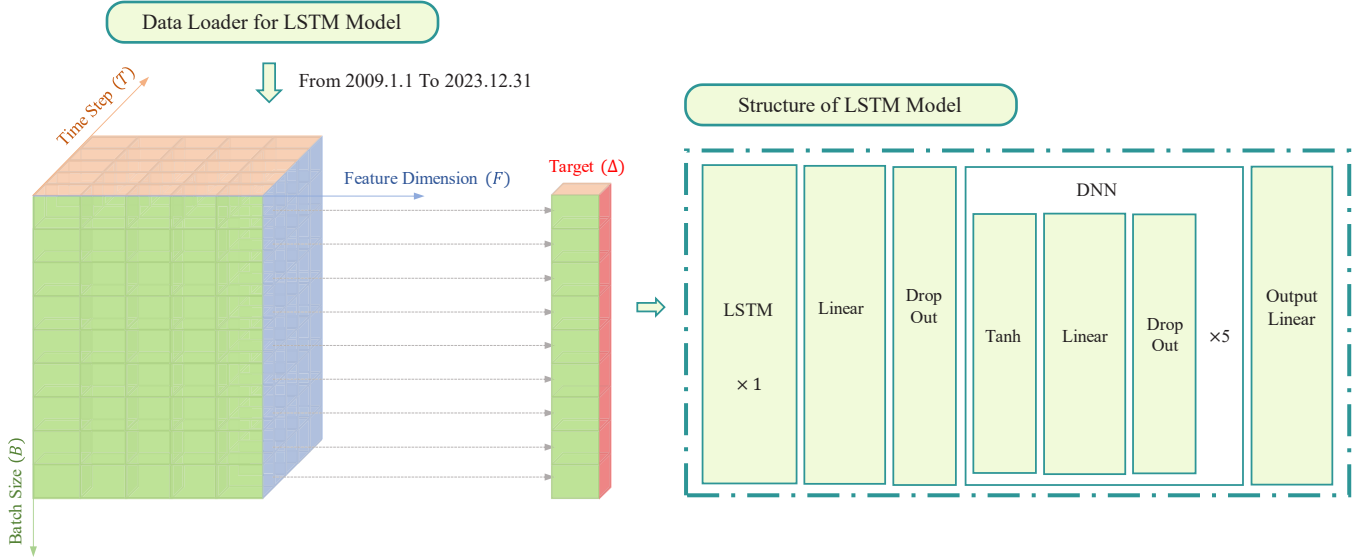


Fig. 22. LSTM Neural Network Structure in This Paper. Note (1) **Tensor Dimensions for Input Data:** In this study, the input data for the LSTM model is represented as a three-dimensional tensor with the shape $[B, T, F]$, capturing the temporal features required for batch processing. The design of this tensor aligns with the structural characteristics of LSTM and the demands of time series forecasting tasks. The specific meanings of each dimension are as follows: a) **Batch Size (B):** This dimension represents the number of samples processed in parallel during training. Batch processing not only enhances the training efficiency but also contributes to the stability of the optimization algorithm. b) **Time Step (T):** This dimension indicates the number of past time steps considered for each sample. In time series modelling, past observations are often crucial for predicting the current state. A fixed-length window of 5 days is used to capture dynamic temporal patterns. c) **Feature Dimension (F):** This dimension denotes the number of input features at each time step. In multivariate time series, each time step include multiple features. By incorporating multidimensional features, the model can analyse the interactions across different dimensions of the time series. (2) **Construction of the Three-Dimensional Tensor:** During data preprocessing, each time series is segmented into sequences of length T using a sliding window approach. These sequences are then stacked in batches to form a three-dimensional tensor. This design satisfies the input format requirements of LSTM while efficiently leveraging GPU parallel computation during training. (3) **Interaction Between Model Architecture and Data Dimensions:** Before entering the LSTM, the input tensor has the shape $[B, T, F]$. After passing through the LSTM, the output tensor undergoes further transformation by flattening the time step dimension: $[B, T, H] \rightarrow [B, T \times H]$, where H represents the hidden layer dimension. This transformation prepares the data for subsequent fully connected layers and prediction tasks. By carefully designing and managing these tensor dimensions, the framework ensures smooth integration between data representation and the model's computational requirements, ultimately enhancing prediction performance.



Fig. 23. Saliency Maps of FXFinBERT-EURUSD-2023 by Google LIT

Appendix B

Table 10
Technical indicators

Category	Variable	Abbrev.	Description
Intra-day Indicators	High vs. Low	HL	The difference between the daily high and low prices, calculated as: High – Low.
	Open vs. Close	OC	The difference between the opening and closing prices for the day, expressed as: Open – Close.
	Low vs. Close	LC	The difference between the daily low and closing prices, determined by: Low – Close.
	High vs. Close	HC	The difference between the daily high and closing prices, given by: High – Close.
Inter-day Indicators	Momentum (1/3/5/10 days)	MM	The price momentum over a specified period (1, 3, 5, or 10 days), calculated as: $Close_t - Close_{t-n}$, where t is the current day and n is the number of days prior.
	Regression Slope (3/5 days)	RS	The slope of the linear regression line fitted to the closing prices over a 3- or 5-day period.
	Moving Average Convergence Divergence	MACD	The difference between the 5-day and 10-day exponential moving averages (EMAs), defined as: $EMA_{5d} - EMA_{10d}$.
	Percentage Price Oscillator	PPO	The percentage difference between the 5-day and 10-day EMAs, calculated as: $\frac{EMA_{5d} - EMA_{10d}}{EMA_{10d}} \times 100$.
	Cross Moving Average (3/5/10 days)	CMA	The intersection points of moving averages calculated over 3, 5, and 10 days, which are indicative of trend changes.
Amplitude Indicators	Momentum Acceleration (3/5 days)	MMA	The change in momentum over a 3- or 5-day period.
	Historical Volatility (3/5/10 days)	VOL	The standard deviation of daily logarithmic returns over a 3-, 5-, or 10-day period, which quantifies price variability.

Table 11
Macroeconomic indicators

Category	Variable	Abbrev.	Description
Equity Market Indicators	S&P 500 Index	GSPC	A benchmark index representing the performance of 500 large U.S. companies.
	NASDAQ Composite Index	IXIC	An index tracking the performance of all stocks listed on the NASDAQ exchange.
	Euronext 100 Index	N100	Tracks the performance of the 100 largest companies listed on the Euronext exchange.
Commodity Market Indicators	EURO STOXX 50 Index	STOXX50E	A blue-chip index tracking the top 50 companies in the Eurozone.
	Gold Price	GOLD	Represents the price of gold, often used as a safe-haven asset indicator.
	WTI Crude Oil Futures	COWTIF	Tracks the price of West Texas Intermediate crude oil futures.
	Bitcoin Price	BIT	Represents the price of Bitcoin as an emerging risk asset.
	CBOE Volatility Index	VIX	Measures market panic or fear by tracking S&P 500 option price volatility.
Bond Market Indicators	CRB Total Return Index	TRCCRBTR	Reflects returns from a broad basket of commodities, including roll yield.
	13-Week Treasury Bill Yield	IRX	Represents short-term U.S. Treasury yields.
	5-Year Treasury Yield	FVX	Measures medium-term U.S. Treasury yields.
	30-Year Treasury Yield	TYX	Captures long-term U.S. Treasury yields.
	Thomson Reuters Eurozone 2-Year Government Benchmark	TRXVEMGOV2E	Tracks 2-year government bond yields in the Eurozone.
	Thomson Reuters Eurozone 5-Year Government Benchmark	TRXVEMGOV5E	Tracks 5-year government bond yields in the Eurozone.
	Thomson Reuters Eurozone 30-Year Government Benchmark	TRXVEMGOV30E	Tracks 30-year government bond yields in the Eurozone.
Inflation Indicators	LIBOR-Treasury Spread	LTS	The difference between the 3-Month LIBOR and the 3-Month Treasury Bill Secondary Market Rate.
	U.S. CPI Y-over-Y Change	USCPIY2Y	Measures annual inflation rate in the U.S. based on CPI.
	U.S. CPI M-over-M Change	USCPIM2M	Measures monthly inflation rate in the U.S. based on CPI.
	Eurozone CPI Y-over-Y Change	EuroCPIY2Y	Measures annual inflation rate in the Eurozone based on CPI.
	Eurozone CPI M-over-M Change	EuroCPIM2M	Measures monthly inflation rate in the Eurozone based on CPI.

Table 12
List of abbreviations

Abbrev.	Full Form
Forex/FX	Foreign Exchange
LLM	Large Language Model
LSTM	Long Short-Term Memory
DDML	Doubly Debiased Machine Learning
RNNs	Recurrent Neural Networks
TI	Technical Indicators
MI	Macroeconomic Indicators
ATE	Average Treatment Effects
KNR	K-Nearest Regressor
ENR	Elastic Net Regressor
SVR	Support Vector Regressor
LGBR	Light Gradient Boosting Regressor
RFR	Random Forest Regressor
GBR	Gradient Boosting Regressor
IR	Improvement Ratio

Table 13
Ablation analysis MAE summary (50 simulations)

Feature Group	Change (%)	Significance	CV (%)
Tech:Intra-day	27.3 %	***	4.1 %
Senti:Change&Continuity	11.2 %	***	6.4 %
Macro:Inflation	10.9 %	***	6.6 %
Macro:Equity	9.9 %	***	7.1 %
Tech:Inter-day	9.6 %	***	7.6 %
Tech:Amplitude	9.5 %	***	5.8 %
Senti:Score	7.3 %	***	6.4 %
Macro:Bond	2.6 %	*	6.6 %
Macro:Commodity	-0.8 %	ns	3.1 %

Table 14
Ablation analysis MSE summary (50 simulations)

Feature Group	Change (%)	Significance	CV (%)
Tech:Intra-day	61.5 %	***	7.4 %
Tech:Inter-day	26.7 %	***	25.5 %
Macro:Equity	26.5 %	***	19.0 %
Macro:Inflation	24.8 %	***	14.1 %
Tech:Amplitude	24.1 %	***	15.4 %
Senti:Change&Continuity	23.1 %	***	16.3 %
Senti:Score	21.5 %	***	17.6 %
Macro:Bond	7.6 %	**	16.8 %
Macro:Commodity	-0.8 %	ns	6.1 %

Table 15
Ablation analysis MSLE summary (50 simulations)

Feature Group	Change (%)	Significance	CV (%)
Tech:Intra-day	61.5 %	***	7.2 %
Macro:Equity	25.4 %	***	18.4 %
Tech:Inter-day	25.2 %	***	24.3 %
Macro:Inflation	24.0 %	***	13.6 %
Tech:Amplitude	23.0 %	***	14.7 %
Senti:Change&Continuity	22.4 %	***	15.8 %
Senti:Score	20.3 %	***	16.9 %
Macro:Bond	6.8 %	**	16.1 %
Macro:Commodity	-0.9 %	ns	6.0 %

Table 16
Ablation analysis MAPE summary (50 simulations)

Feature Group	Change (%)	Significance	CV (%)
Tech:Intra-day	27.3 %	***	4.0 %
Senti:Change&Continuity	11.1 %	***	6.2 %
Macro:Inflation	10.7 %	***	6.4 %
Macro:Equity	9.5 %	***	6.9 %
Tech:Inter-day	9.2 %	***	7.3 %
Tech:Amplitude	9.2 %	***	5.6 %
Senti:Score	7.0 %	***	6.2 %
Macro:Bond	2.3 %	*	6.4 %
Macro:Commodity	-0.8 %	ns	3.0 %

Table 17
Ablation analysis LogCosh summary (50 simulations)

Feature Group	Change (%)	Significance	CV (%)
Tech:Intra-day	61.5 %	***	7.4 %
Tech:Inter-day	26.7 %	***	25.5 %
Macro:Equity	26.5 %	***	19.0 %
Macro:Inflation	24.8 %	***	14.1 %
Tech:Amplitude	24.1 %	***	15.4 %
Senti:Change&Continuity	23.1 %	***	16.3 %
Senti:Score	21.5 %	***	17.6 %
Macro:Bond	7.6 %	**	16.8 %
Macro:Commodity	-0.8 %	ns	6.1 %

Appendix C

Prompt of LLM

Classify the sentiment of the following news related to EUR/USD=X exchange rate as positive, negative, or neutral based on whether it likely causes the rate to rise, fall, or has no direct effect. Ensure the output is strictly one of the following: positive, negative, or neutral.

Data availability

Data will be made available on request.

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